



PHYSICS MODULE 7

The Nature of Light



PHYSICS MODULE 7: THE NATURE OF LIGHT

ELECTROMAGNETIC SPECTRUM

Inquiry question: What is light?

- investigate Maxwell's contribution to the classical theory of electromagnetism, including:
 - unification of electricity and magnetism
 - prediction of electromagnetic waves
 - prediction of velocity (ACSPH113)  
- describe the production and propagation of electromagnetic waves and relate these processes qualitatively to the predictions made by Maxwell's electromagnetic theory (ACSPH112, ACSPH113)
- conduct investigations of historical and contemporary methods used to determine the speed of light and its current relationship to the measurement of time and distance (ACSPH082)  
- conduct an investigation to examine a variety of spectra produced by discharge tubes, reflected sunlight or incandescent filaments
- investigate how spectroscopy can be used to provide information about: 
 - the identification of elements
- investigate how the spectra of stars can provide information on:  
 - surface temperature
 - rotational and translational velocity
 - density
 - chemical composition

LIGHT: WAVE MODEL

Inquiry question: What evidence supports the classical wave model of light and what predictions can be made using this model?

- conduct investigations to analyse qualitatively the diffraction of light (ACSPH048, ACSPH076) 
- conduct investigations to analyse quantitatively the interference of light using double slit apparatus and diffraction gratings $d \sin \theta = m \lambda$ (ACSPH116, ACSPH117, ACSPH140)  
- analyse the experimental evidence that supported the models of light that were proposed by Newton and Huygens (ACSPH050, ACSPH118, ACSPH123) 
- conduct investigations quantitatively using the relationship of Malus' Law $I = I_{\max} \cos^2 \theta$ for plane polarisation of light, to evaluate the significance of polarisation in developing a model for light (ACSPH050, ACSPH076, ACSPH120)  

LIGHT: QUANTUM MODEL

Inquiry question: What evidence supports the particle model of light and what are the implications of this evidence for the development of the quantum model of light?

- analyse the experimental evidence gathered about black body radiation, including Wien's Law related to Planck's contribution to a changed model of light (ACSPH137)   
 - $\lambda_{\max} = \frac{b}{T}$

- investigate the evidence from photoelectric effect investigations that demonstrated inconsistency with the wave model for light (ACSPH087, ACSPH123, ACSPH137) 
- analyse the photoelectric effect $K_{\max} = hf - \phi$ as it occurs in metallic elements by applying the law of conservation of energy and the photon model of light, (ACSPH119) 

LIGHT AND SPECIAL RELATIVITY

Inquiry question: How does the behaviour of light affect concepts of time, space and matter?

- analyse and evaluate the evidence confirming or denying Einstein's two postulates:
 - the speed of light in a vacuum is an absolute constant
 - all inertial frames of reference are equivalent (ACSPH131)
- investigate the evidence, from Einstein's thought experiments and subsequent experimental validation, for time dilation $t = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$ and length contraction $l = l_0 \sqrt{1 - \frac{v^2}{c^2}}$, and analyse quantitatively situations in which these are observed, for example:
 - observations of cosmic-origin muons at the Earth's surface 
 - atomic clocks (Hafele–Keating experiment) 
 - evidence from particle accelerators 
 - evidence from cosmological studies 
- describe the consequences and applications of relativistic momentum with reference to:
 - $p_v = \frac{m_0 v}{\sqrt{1 - \frac{v^2}{c^2}}}$ 
 - the limitation on the maximum velocity of a particle imposed by special relativity (ACSPH133) 
- Use Einstein's mass–energy equivalence relationship $E = mc^2$ to calculate the energy released by processes in which mass is converted to energy, for example: (ACSPH134) 
 - production of energy by the sun
 - particle–antiparticle interactions, eg positron–electron annihilation
 - combustion of conventional fuel

HUYGENS WAVE MODEL OF LIGHT

OUTCOMES

- conduct investigations to analyse qualitatively the diffraction of light (ACSPH048, ACSPH076) 🖥️
- conduct investigations to analyse quantitatively the interference of light using double slit apparatus and diffraction gratings $d\sin\theta = m\lambda$ (ACSPH116, ACSPH117, ACSPH140) 🖥️ 📐
- analyse the experimental evidence that supported the models of light that were proposed by Newton and Huygens (ACSPH050, ACSPH118, ACSPH123) ⚙️
- conduct investigations quantitatively using the relationship of Malus' Law $I = I_{\max}\cos^2\theta$ for plane polarisation of light, to evaluate the significance of polarisation in developing a model for light (ACSPH050, ACSPH076, ACSPH120) 🖥️ 📐

EARLY CLASSICAL MODELS OF LIGHT (HUYGENS AND NEWTON)

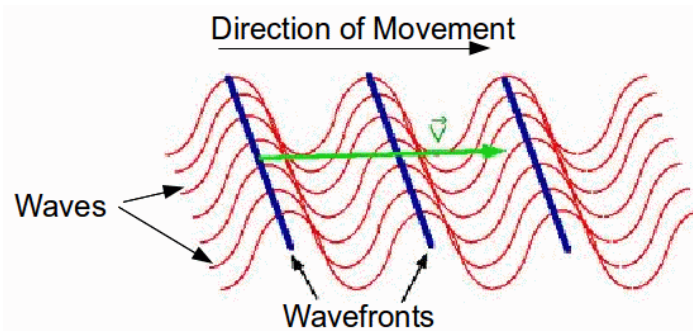
The main two classical theories regarding the nature of light were by **Christian Huygens** and **Isaac Newton**.

- **Huygens** proposed that light was a **wave**
- **Newton** proposed that light was made out of **particles**

HUYGENS' WAVE MODEL

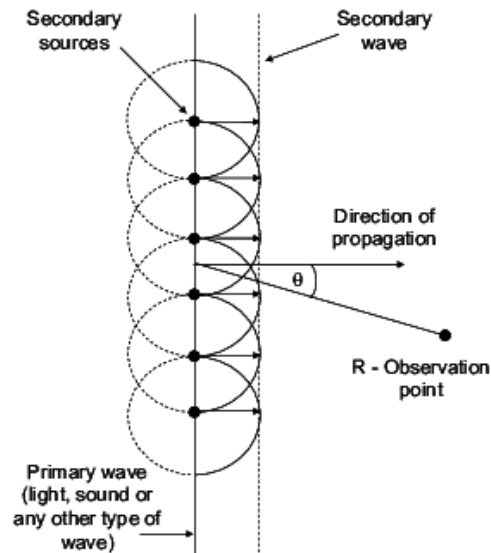
The propagation of a wave can be represented using **wavefronts** and **rays**.

- **Wavefronts** represent places of constant phase. E.g. the **crests** of the wave. Therefore, the **distance between successive wavefronts is the wavelength λ** .
- The **ray** represents the **direction of propagation**.
- The **ray is perpendicular to wavefronts**.



The **main features** of Huygens' Model are:

- Huygens proposed that **space** is filled with **small elastic particles** ("Ether")
- Light is a **longitudinal wave through these perfectly elastic particles that fill space** (This would be similar to sound travelling in air). No kinetic energy is lost during collisions
- Every part of the wave is a **new source of a new wave** (called a **wavelet**). This is called **Huygens' Principle**.



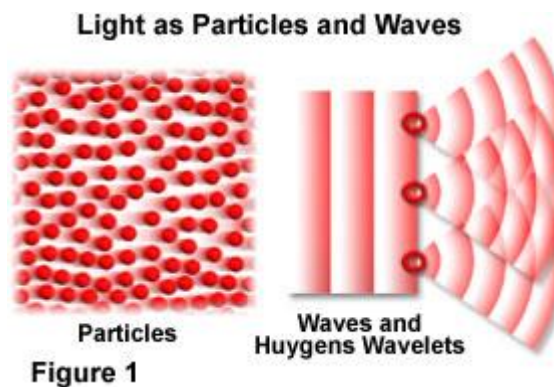
Huygens proposed that all **wavelets superimpose** and **interfere** to produce the **wavefront** of the propagating wave.

NEWTON'S PARTICLE MODEL

The main features of Newton's model are:

- Light consists of **corpuscles** (small **particles** with **mass**)
- The corpuscles are **lower in mass and elastic**
- Corpuscles of **different sizes** (and different masses) correspond to light of **different colours**.

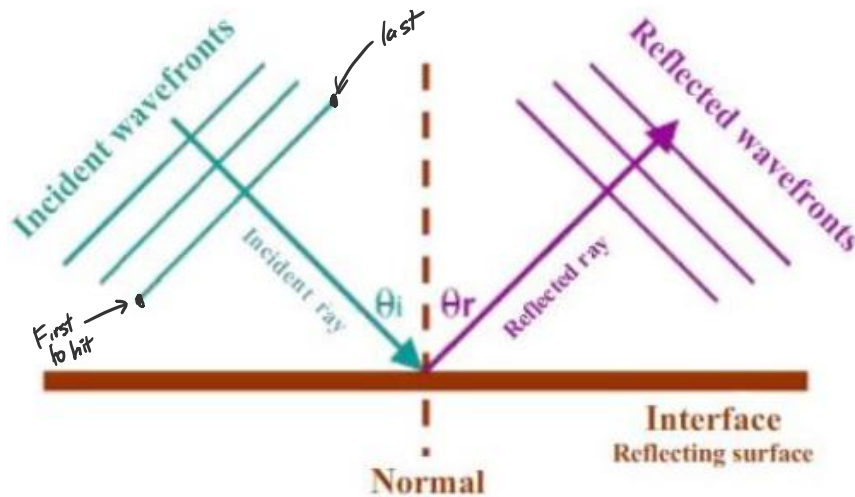
Newton attempted to explain the behaviour of light using **mechanics**: the effect of forces acting on small particles.



INITIAL EVIDENCE FOR EARLY CLASSICAL MODELS

REFLECTION OF LIGHT

Light was known to reflect off a smooth surface (like a mirror) following the **law of reflection**, where the angle of incidence is equal to the angle of reflection ($\theta_i = \theta_r$).

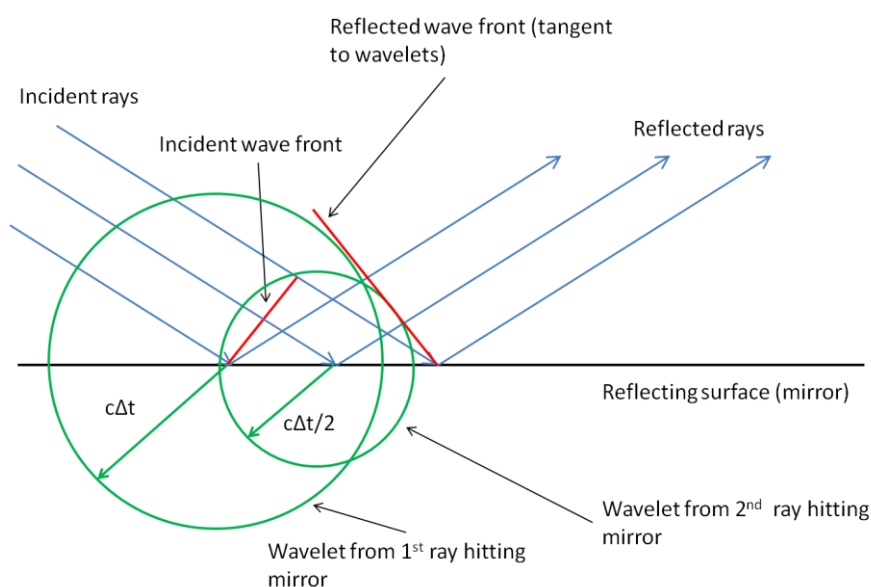


HUYGENS' EXPLANATION OF REFLECTION

Huygens explained the reflection of light using his **Huygens' Principle** – through the formation of wavelets.

- The wavefront hitting a surface would produce wavelets, however these cannot propagate forward as the surface blocks them.
- The wavelets spread out from the surface and form the reflected wavefront.

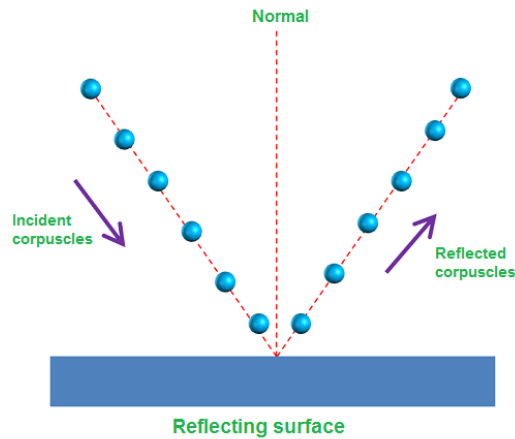
Huygens' Principle for Reflection: Correct Way



NEWTON'S EXPLANATION OF REFLECTION

Newton's explained the reflection of light as the **reflection of particles** off a surface.

- He proposed that the collision was an **elastic collision** (no loss of kinetic energy, so the speed of the corpuscles was the same before and after a collision),
- He considered the force acting on the corpuscles and conservation of energy.

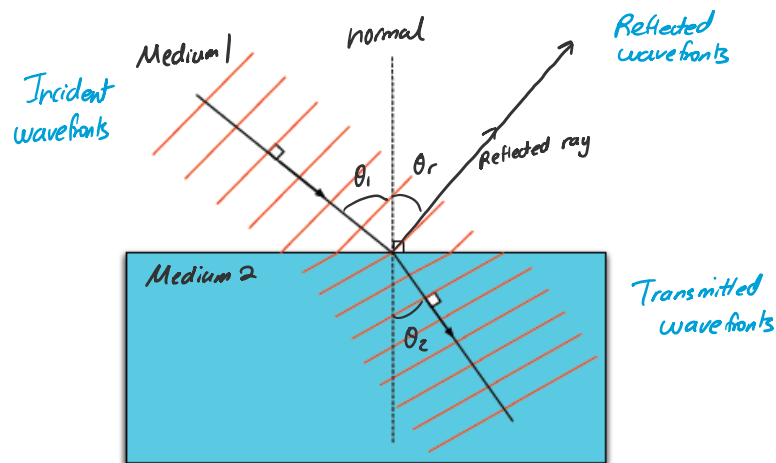


SUMMARY – REFLECTION

Both models predict that a ray will be reflected at an angle equal to its angle of incidence. They are both consistent with the **law of reflection**.

REFRACTION OF LIGHT

Refraction describes a **wave bending** as it **moves from one medium to another**. It follows the **Principle of Least Time**, where a light ray will take the shortest path of time.



Refraction occurs as the **speed of the wave** changes when it **enters a new medium**. Refraction is described by **Snell's Law**, which relates the incidence angle (θ_1) to the refracted angle (θ_2).

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{v_1}{v_2} \text{ or } n_1 \sin \theta_1 = n_2 \sin \theta_2$$

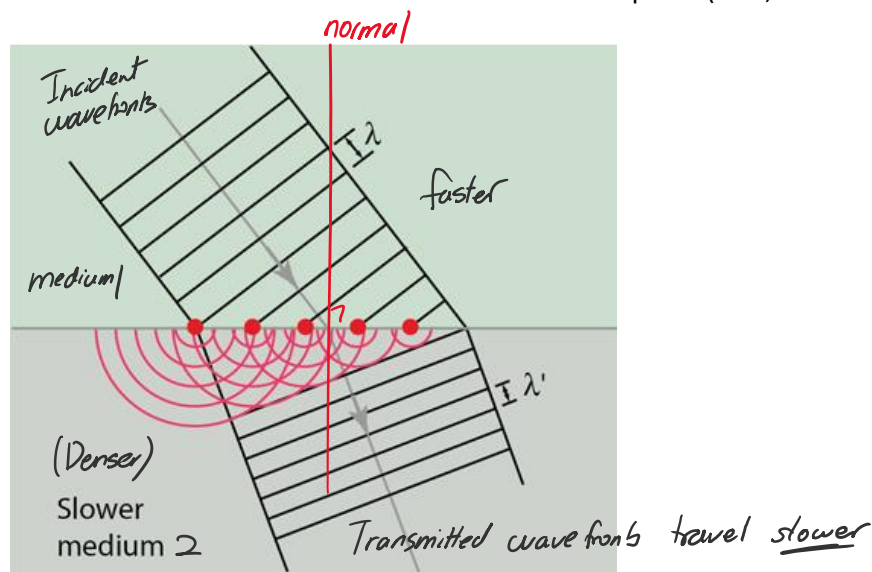
- v_1 and v_2 are the velocities of the light in medium 1 and 2 respectively.
- n_1 and n_2 are the refractive indices of medium 1 and 2 respectively.
- The angle of incidence and refraction are measured **relative to the normal**.

The speed of light in different materials was not known at the time of Newton and Huygens.

HUYGENS' EXPLANATION OF REFRACTION

Huygens explained that refraction in terms of **velocity changing**. He proposed that the **wavefronts slow down when they enter a denser medium**,

- By slowing down, the **wavefronts bend towards the interface**.
- Hence, the **ray bends towards the normal in a denser medium**.
- The wavelength must reduce in the denser medium due to the slower wave speed. (Also, the wave equation)



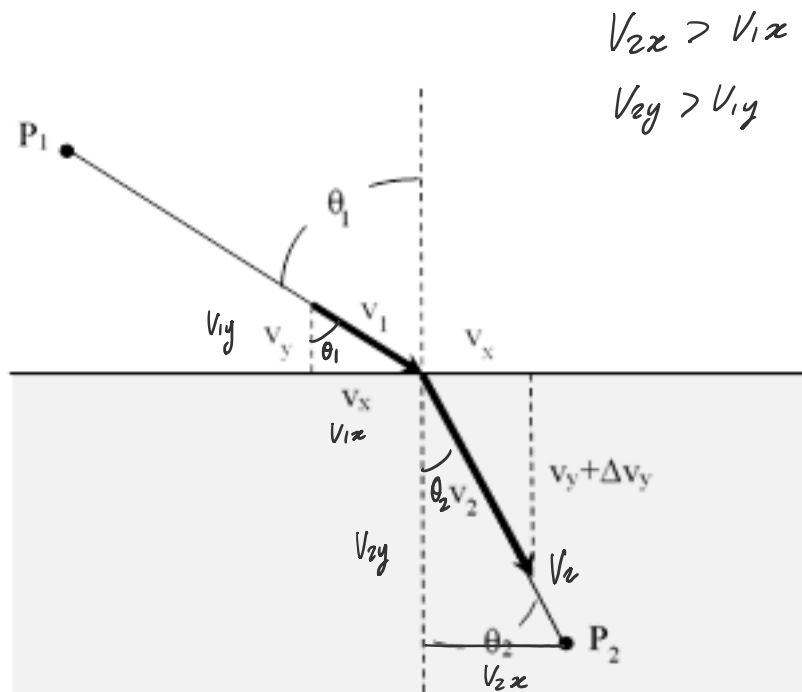
In 1678 Huygens' used this model to derive Snell's Law: $\frac{\sin \theta_1}{\sin \theta_2} = \frac{v_1}{v_2}$.

Different colours refract by different amounts (e.g. violet refracts more than red), known as **dispersion**. Huygens' model explains this by saying the different colours (different λ) have different speeds in the denser medium.

NEWTON'S EXPLANATION OF REFRACTION

Newton proposed refraction in terms of **forces acting on the corpuscles**.

- He proposed that corpuscles interact with the material they travel through and experience a force.
- If the corpuscles are in **uniform medium (constant density)**, that force is equal in all directions, making it a **net zero force**.
- The denser medium contains more matter and can **impart a stronger force** on the corpuscles. As the corpuscles approach the **denser medium**, there will be a **net force towards it**.
 - o The corpuscles will accelerate towards the denser medium (at the surface of the interface)
- The corpuscles' velocity **increases perpendicular to the interface** and remains **constant parallel to the interface**. Hence the corpuscles changes direction.



Newton's version of Snell's Law: $\frac{\sin \theta_1}{\sin \theta_2} = \frac{\frac{v_{1x}}{v_1}}{\frac{v_{2x}}{v_2}} = \frac{v_2}{v_1}$. Newton is assuming that light will **speed up** in a **denser medium**. The light will bend **towards the normal** in the denser medium. This means that the dispersion of violet light refracts more than red light. Newton argued that:

- All corpuscles experienced the **same force** as they moved from one medium to another.
- The **red** corpuscles were more massive and had **more inertia**. Hence, they refracted less than the violet corpuscles which had less inertia.

SUMMARY – REFRACTION

Both Huygens' and Newton's model could explain refraction. They both predict that light would bend towards the normal when it enters a denser medium. However, Newton claimed that the light will speed up in the denser medium.

More accurate measurements of the speed of light is needed to differentiate between the two different models. Although the experimental evidence supported Huygens' wave model, Newton had been an incredibly successful and influential scientist and people continued to support his model because of his reputation for approximately 100 years after his death.

DIFFRACTION

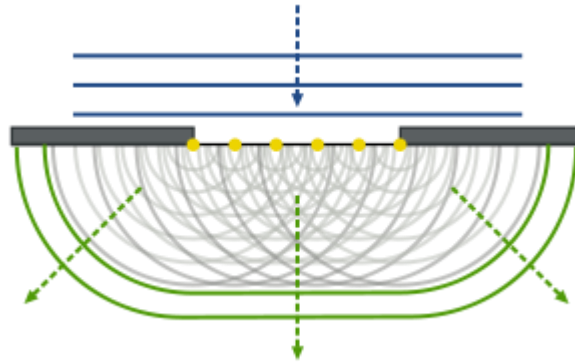
Diffraction is the **bending of waves** around an **obstacle**. If a wave encounters a barrier that has an opening of dimensions similar to its wavelength, the wave will flare out into the region beyond the barrier.

Diffraction is only obvious when the **wavelength is comparable to the size of the slit or object** that causes the diffraction.

EXPLANATION OF DIFFRACTION USING HUYGENS' PRINCIPLE

The diffraction of light was explained by August-Jean Fresnel who said:

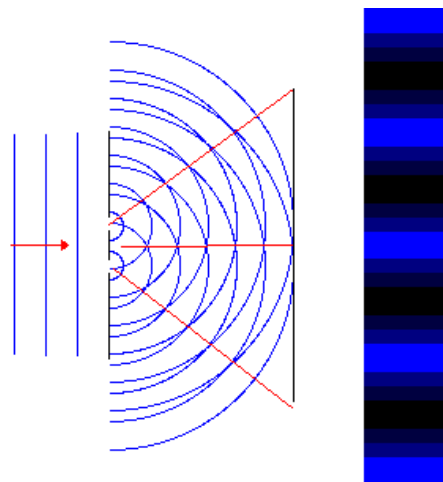
- The wavelets **away** from the barrier **interfere and cancel out in all directions** other than the forward direction and produce a straight propagating wavefront.
- The wavelets hitting the barrier are blocked and produced no waves.
- The wavelets near the edge only interact with waves on one side and interfere to produce the straight wavefront. On the other side, they are **free to propagate**, and produce **curved wavefronts**.



Newton did not provide an explanation for diffraction.

INTERFERENCE

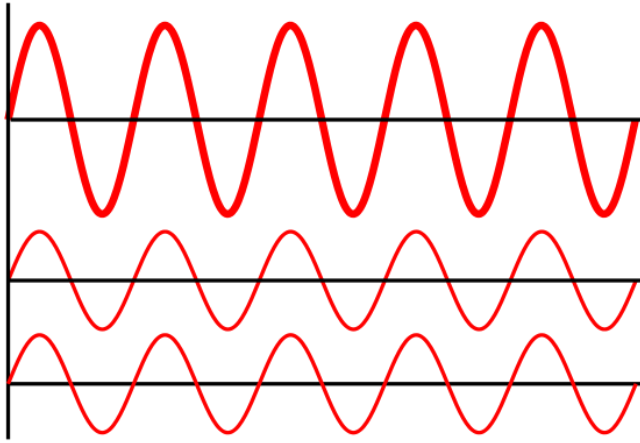
Diffraction leads to wave spreading out. If the waves spread out and end up travelling over the same space, the waves become **superimposed**. The waves **overlap** and interfere with each other producing a pattern.



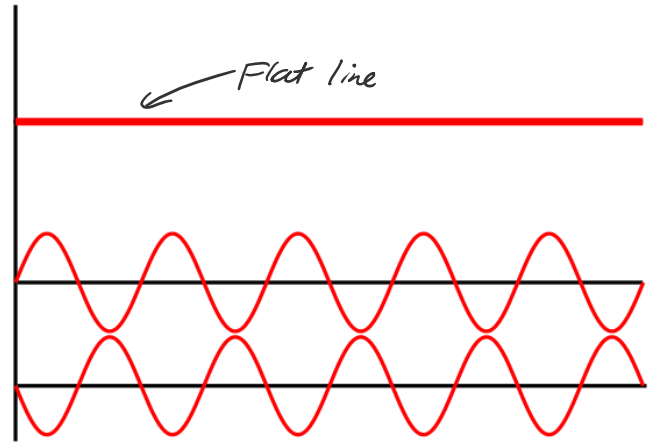
Depending on the **relative phase** of superimposed waves, waves can undergo **constructive** and **destructive** interference.

- If the waves **add in phase**, they **constructively interfere** resulting in a **larger amplitude** (or **brighter** region of light).
- If the waves **add out of phase**, they **destructively interfere** resulting in a **smaller amplitude** (or **darker** region of light).

Constructive Interference



Destructive Interference

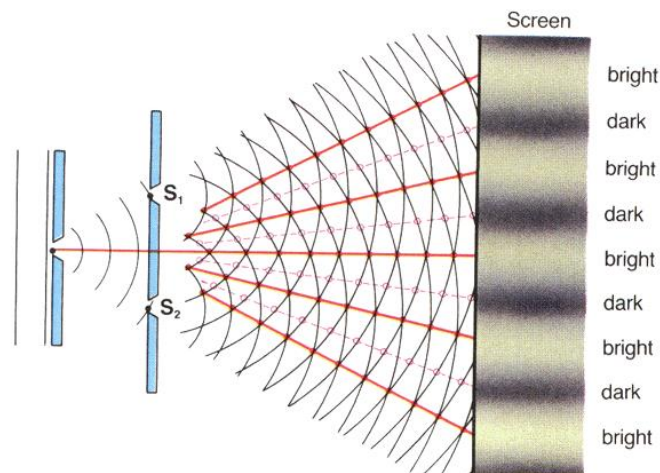


THOMAS YOUNG'S DOUBLE SLIT EXPERIMENT

Thomas Young devised an experiment in 1801 to demonstrate that **light** is a **wave**. He showed that light undergoes **diffraction** and **interference** - both phenomena associated with waves.

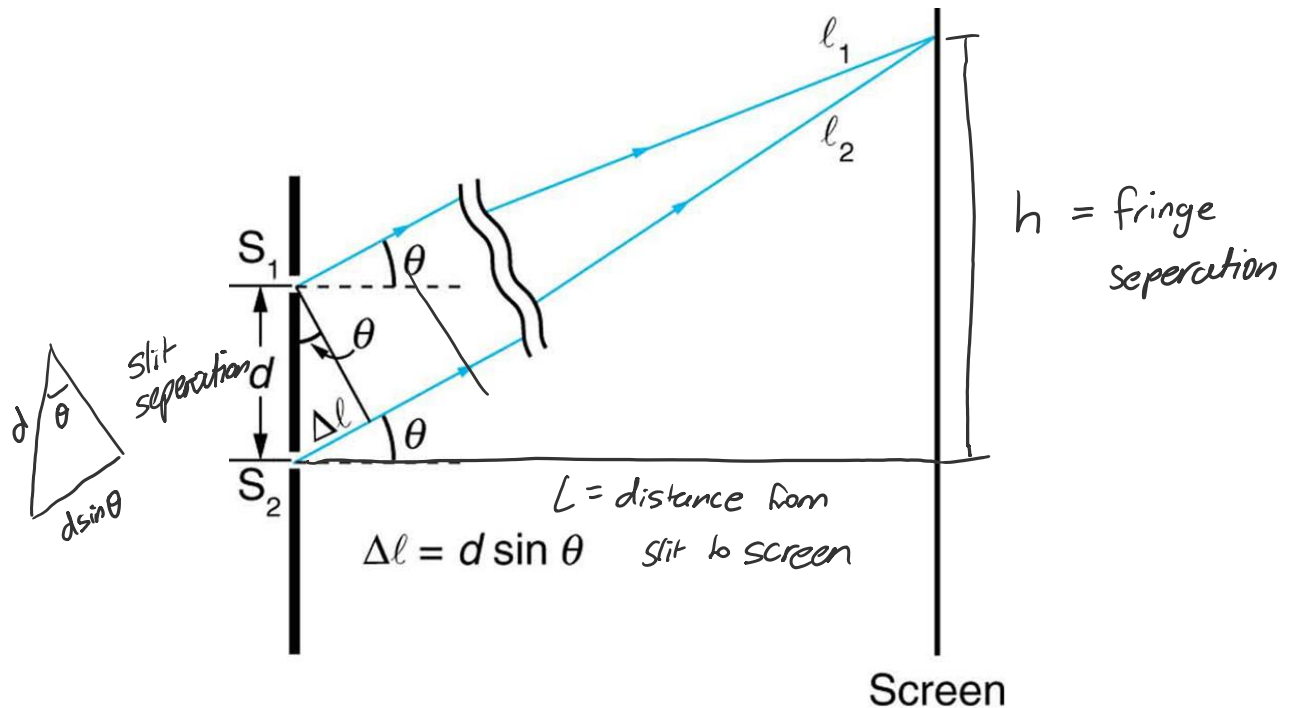
He allowed light to pass through a **double slit** barrier. This resulted in the formation of an **interference pattern** consisting of **dark** and **bright fringes** on an observation screen.

Interference Patterns



Each slit diffracts the incident light and emits circular wavefronts that overlap and interfere with those from the other slit. Where the rays from each slit arrive at the screen in phase, they interfere constructively and produce a bright spot (fringe). Where the waves are out of phase, they interfere destructively and produce a dark spot.

Consider light that travels at an angle of θ to the horizontal to reach a distance screen as shown in the diagram below:



- If the screen is far away from the slits it can be assumed that the paths of light from the two slits are **parallel**
- The path difference is given by **$d \sin \theta$** where d is the **distance between the slits**.
- Hence **constructive interference** will occur for angles that satisfy

$$d \sin \theta = m\lambda \quad m = 0, \pm 1, \pm 2, \pm 3 \dots$$

- If θ is a small angle (for $h \ll L$) we can assume that:

$$\sin \theta \approx \tan \theta = \frac{h}{L}$$

$$d \left(\frac{h}{L} \right) = m\lambda$$

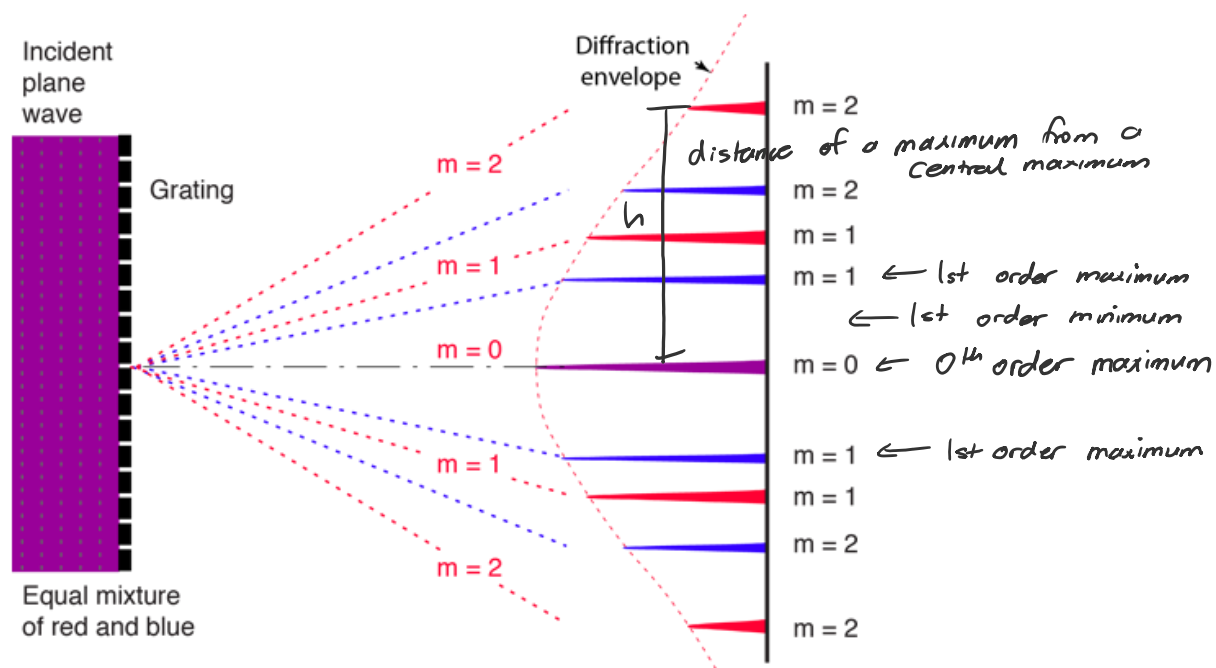
Using the equations, the wavelength can be calculated if the distance between the slits and the angle to the bright fringes is measured.

- Apart from inferring that light has wave properties from the fact that it diffracts and interferes, this allows the **wavelength to be measured** directly.
- Young's results showed that light of **different wavelengths** corresponds to **different colours**.

DIFFRACTION GRATINGS

A diffraction grating is a very useful device for **analysing light sources** (e.g. **determining wavelength**) and consists of a large number of equally spaced parallel slits, a distance d apart.

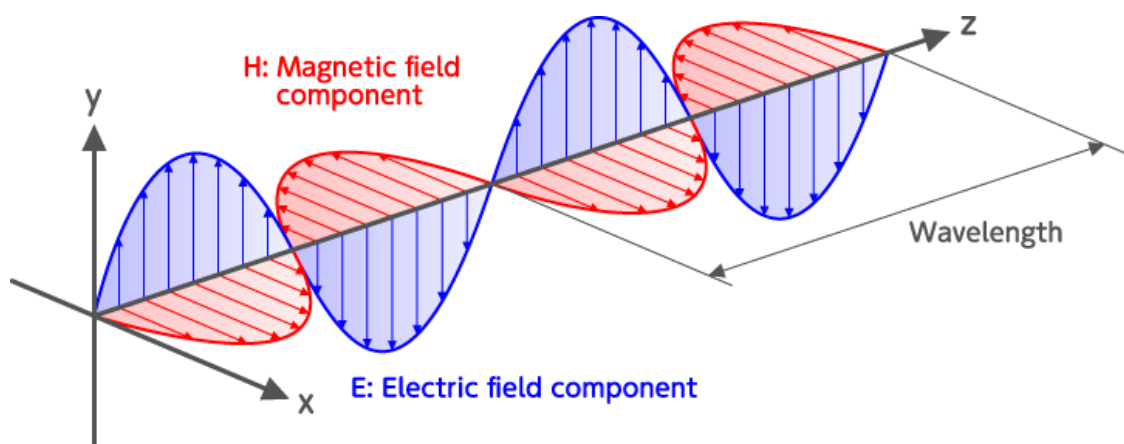
- A grating can be made by **engraving parallel lines** on a **glass plate**. The spaces between each line are transparent and hence act as separate slits.
- A typical grating consists of **several thousand lines per centimetre**.



- Light passing through each slit diffracts and interferes and will produce a diffraction pattern of bright spots on a screen.
- If the screen is **far away** from the grating, we can assume that the rays converging on a certain point on the screen are **almost parallel**.

The condition for a grating is the same as the condition for the double slit experiment. Therefore, the same equations can be used to calculate wavelength.

POLARISATION AND TRANSVERSE WAVE NATURE



Polarisation refers to the **direction** in which the **electric field oscillates**.

- If the electric field **oscillates vertically**, it is **vertically polarised**.
- If the electric field **oscillates horizontally**, it is **horizontally polarised**.

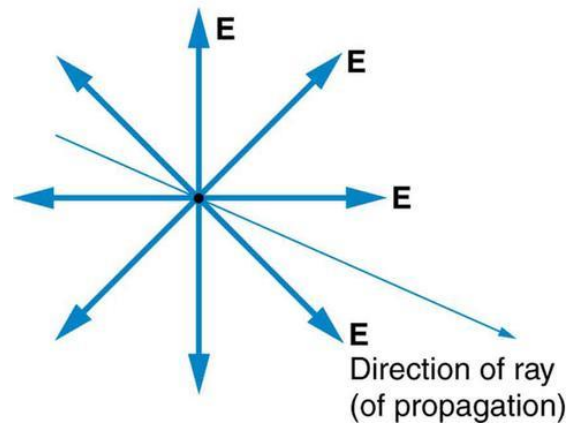
The **electric field** (and magnetic field) can oscillate in any direction as long as they **remain perpendicular** to each other and to the direction of propagation.

- An electromagnetic wave's polarisation defines the **axis** along which the **electric field oscillates**.

An ordinary beam of **visible light** is **unpolarised**, meaning the electric field will oscillate along different axes.

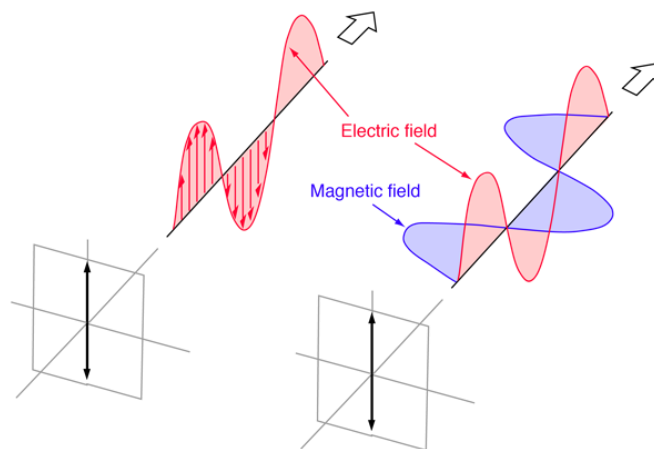
- Waves are emitted by a large number of molecules of the light source, with each wave having its own orientation of electric field oscillation
- All waves mix together at slightly different positions to give an **unpolarised beam of light**

Random polarization



Representation of a mixture of electric field oscillations (Not net)

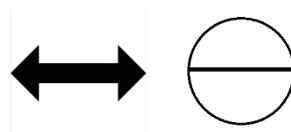
However, the oscillations of transverse waves may be **restricted** to a **single plane**. In this case, the wave is **linearly polarised**. The electric field oscillates along the same axis at all times at a particular point.



POLARISERS

A **polariser**, polaroid, or polarising filter is a material that allows for only **one polarisation of light** to be **transmitted** (the other is either reflected or absorbed).

- It is typically represented by a circle with a line representing the **polarisation** that is **transmitted**.



- If light is polarised **perpendicular** to the polariser, it will **not be transmitted**.
- If light is **unpolarised**, only **some light will be transmitted**, and the transmitted light will be **polarised**.

MALUS'S LAW

Malus's law was discovered by Etienne-Louis Malus in 1809 and quantifies the extent to which a **polariser** allows a **polarised beam** of light to be **transmitted**. The transmitted light intensity is given by:

$$I = I_{\max} \cos^2 \theta$$

Where:

- $I_{\max}$: incident intensity
- θ : Angle between the incident polarisation and the polariser

ELECTROMAGNETIC WAVES

OUTCOMES

- investigate Maxwell's contribution to the classical theory of electromagnetism, including:
 - unification of electricity and magnetism
 - prediction of electromagnetic waves
 - prediction of velocity (ACSPH113) ⚙️💻
- describe the production and propagation of electromagnetic waves and relate these processes qualitatively to the predictions made by Maxwell's electromagnetic theory (ACSPH112, ACSPH113)
- conduct investigations of historical and contemporary methods used to determine the speed of light and its current relationship to the measurement of time and distance (ACSPH082) ⚙️💻

MAXWELL'S EQUATIONS

- Gauss's Law
 - Charges produce electric fields that diverge
- Gauss's Law for magnetism
 - Magnetic fields do not diverge, they form closed loops
- Faraday's Law
 - A changing magnetic field produces an EMF that curls around into a circle.
- Ampere's Law with Maxwell's addition
 - A current and/or a changing electric field produces a magnetic field that curls around into a circle.

The significance of Maxwell's work on his equations is that they represent the **unification of electricity and magnetism**.

- Maxwell showed that electricity and magnetism cannot be expressed using separate equations because of the way electric and magnetic fields affect each other
- Faraday's Law of induction shows that **electric fields are produced by changing magnetic fields**.
- Ampere's law shows that **magnetic fields are produced by** currents and by **changing magnetic fields**
- Hence, electricity and magnetism are two aspects of one phenomenon: **electromagnetism**.

ELECTROMAGNETIC WAVES

Maxwell combined his equations describing electromagnetism in different ways to produce another equation called the **wave equation**.

- The form of the wave equation was already known to describe other waves.
- Hence, Maxwell proposed that waves of electric and magnetic fields must be possible. He **predicted** the existence of **electromagnetic waves**
 - Electromagnetic waves consisted of **oscillating electric and magnetic fields**.

The wave equation also gives you the **speed of the wave**. Maxwell showed that the speed of electromagnetic waves was $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$. This discovery led Maxwell to **propose** that **visible light is an electromagnetic wave**. His proposal consisted of two parts:

- 1) Electromagnetic waves consist of self-sustaining mutual generation of electric and magnetic fields that propagate at the speed of light. Electric fields create magnetic fields, and magnetic fields create electric fields, resulting in a propagating wave.
- 2) Visible light is an electromagnetic wave or electromagnetic oscillation.

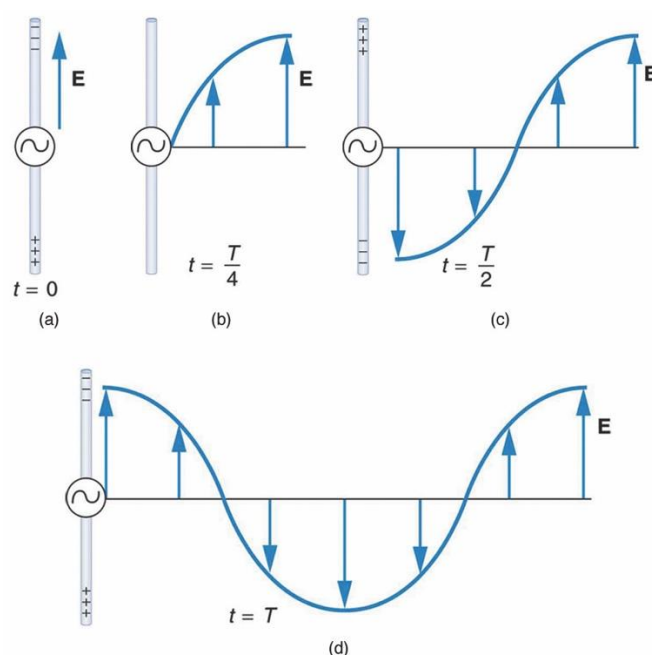
PRODUCTION OF ELECTROMAGNETIC WAVES

Electromagnetic waves can be generated by **accelerating a charge**. It changes both its magnetic and electric fields in the space surrounding it. These changes then travel away from the charge as an electromagnetic wave. The oscillations of each field, sustain those of the other. (According to Maxwell's formulation of Faraday's and Ampere's laws).

AC SUPPLY & ELECTROMAGNETIC WAVES

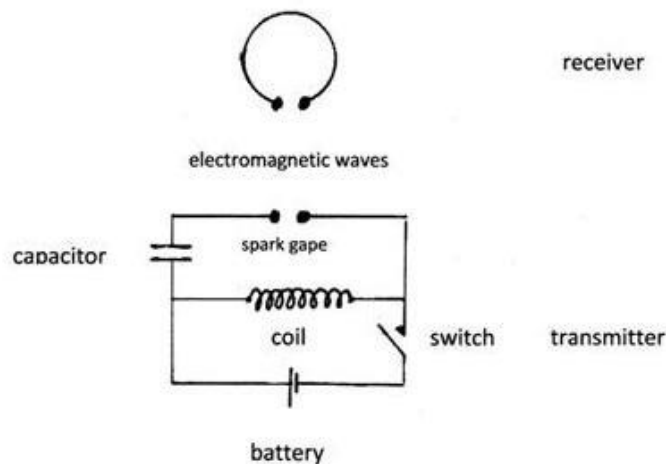
An AC voltage applied to the wires of an antenna forces electric charges in the antenna to **oscillate**. This is a common technique for **accelerating charged particles** and is the source of radio waves emitted by the antenna of a radio station.

Two metal rods that are connected to an AC generator will produce a sinusoidal output voltage. This voltage causes charges to oscillate between the two ends of the rods



- The electric field near the antenna oscillates in phase with the charge distribution. The field points down when the upper rod is positive, and up when the upper rod is negative. The **magnitude** of the field at any instant depends on the amount of charge in the rod at that instant.
 - o The **frequency** of the electric field oscillations is the same as the **AC input**, and the **amplitude** of the electric field oscillations depends on the **amplitude of the AC input**.
- As the charges continue to oscillate and accelerate between the rods, the electric field set up by the charges move away from the antenna at the speed of light
 - o **One cycle of charge oscillations** produces **one wavelength** in the electric field oscillation
 - o This process creates **linearly polarised** wave, because the electric field is oscillating along a line.

Hertz showed that **the nature of reflection, refraction and polarisation** of radio waves was the same as those of **visible light** and confirmed that the nature of electromagnetic waves was as Maxwell predicted, and that visible light is an electromagnetic wave.



The **emitter antenna** is connected to a power supply. When connected, sparks (accelerating charges) form in a gap at the centre of the antenna which indicate that current flows in the antenna. This generates the radio waves.

Hertz stood some distance away, holding the receiver antenna, which also had a gap in it. Sparks across the gap in the receiver indicate a current flowing in the antenna. However, since it is not connected to a power supply, this antenna must be detecting radio waves.

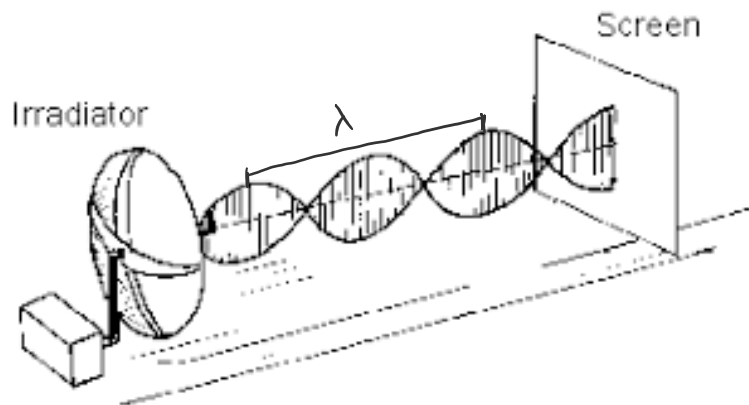
Feature	Description	Purpose
Induction coil	It provides short high voltage surges to the electrodes at the gap of the antenna	Short high voltages between the electrodes ionises air, resulting in sparks. This allows charges to accelerate up and down the antenna as AC flows through the antenna.
Electrodes	Brass knobs capable of carrying a large amount of positive and negative charges	When a large potential difference (voltage) between the electrodes is created, it allows the charges to be built up to move across the gap and make sparks.
Parabolic metal mirrors	Reflects the electromagnetic waves produced by the transmitter	To focus the weak radio waves on the receiver
Receiver	A loop of wire connected to the electrodes with its own spark gap	To receive and detect the radio waves produced by the transmitter. The receiver is tuned to the same frequency as the transmitter.

HERTZ'S INVESTIGATION OF EM WAVES

Hertz calculated the speed of light to be $3 \times 10^8 \text{ m/s}$. He also showed that the radiation generated by the sparks exhibited **wave properties** and concluded that radio-frequency waves had properties similar to visible light.

REFLECTION & INTERFERENCE

Hertz reflected the generated waves off a metal sheet. He set up standing waves.



A standing wave, also known as a stationary wave, is formed when two identical waves travel in opposite directions interfere and result in constructive and destructive interference.

SPEED OF ELECTROMAGNETIC WAVES

In order to calculate the speed of the waves, the frequency and wavelength of the wave must be known.

- Hertz calculated the **frequency of oscillations** in the transmitter
- He then formed a **standing wave** by reflecting the radio waves off a large flat zinc plate. He moved his receiver coil along this wave. A spark in the gap was produced at **antinodes** (areas of constructive interference) but not at the **nodes** (areas of destructive interference). The distance travelled between two nodes and antinodes is half wavelength.

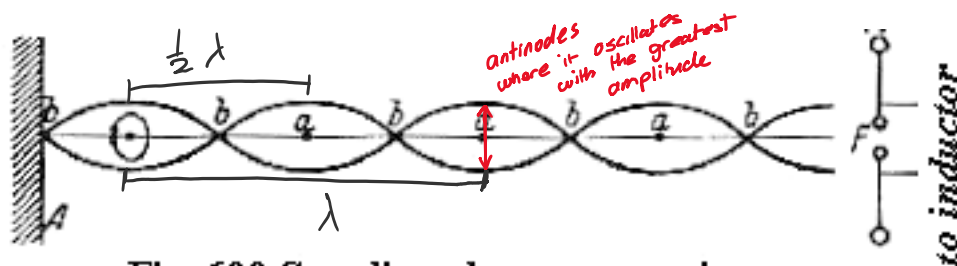


Fig. 600 Standing electromagnetic wave

- He was able to calculate the wavelength and then the speed of light. His value confirmed Maxwell's prediction.

POLARISATION OF ELECTROMAGNETIC WAVES

- He discovered that a spark would only be induced in the detecting loop if the gap in the detecting loop was aligned **parallel** to the gap in the transmitter loop.
- If the detecting loop was perpendicular to the transmitter loop, no spark could be induced in the detecting loop.

The reason for the orientation of the detector being important is that:

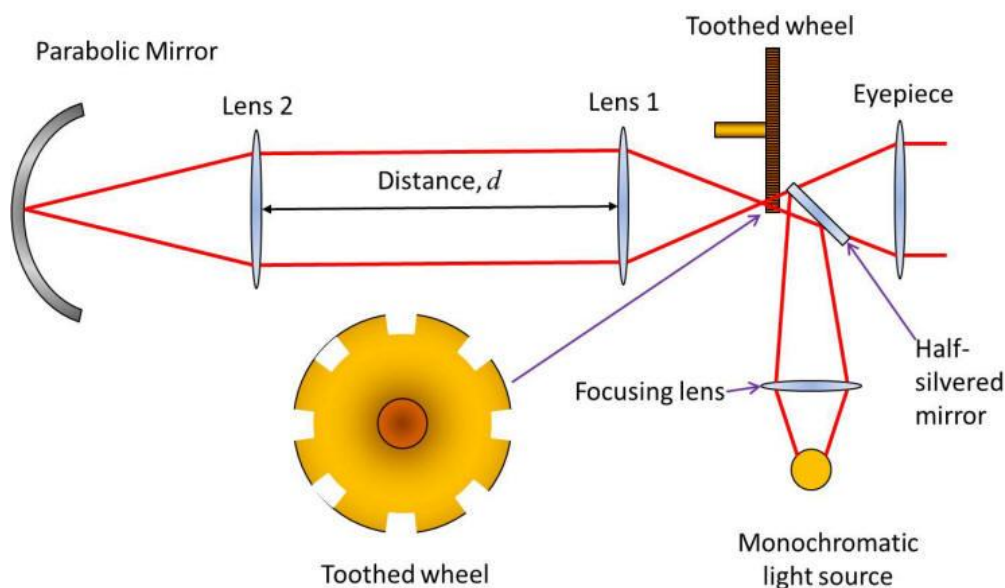
- The electric field in the transmitter oscillates along **one axis**. This produces **polarised waves** in this direction.
- The receiver can only detect waves with an electric field oscillating across the spark gap, so is only sensitive to radio waves polarised in this direction

TIME OF FLIGHT

Time of flight measurements rely on timing how long it takes for light to travel a path of known length and calculating its speed using $v = \frac{s}{t}$.

In 1849, Fizeau reflected a beam of light from a mirror 8 km away such that it returned to a target. He placed a **rotating cogwheel** in the path of the light.

- The light had to go through one of the gaps between the cogs and return in the following gap. If it does not and the returning light hits one of the teeth, no light is observed. The angular spin of the cog is adjusted such that light can be seen returning.



- Fizeau measured the speed of light as $3.15 \times 10^8 \text{ m/s}$.

USES OF ELECTROMAGNETIC WAVES

TABLE 9.1.1 A comparison of the different waves in the electromagnetic spectrum and their effects on matter.

Type of wave	Typical wavelength (m)	Typical frequency (Hz)	Comparable size	Effect on matter
AM radio wave	100	3×10^6	sports oval	causes movement of free electrons in a conductor
FM radio or TV wave	3	1×10^8	small car	
microwaves	0.03	1×10^{10}	50c coin	causes molecular rotation
infrared	10^{-5}	3×10^{13}	white blood cell	makes chemical bonds vibrate
visible light	10^{-7}	3×10^{15}	small cell	affects electronic states in atoms or chemical bonds
ultraviolet	10^{-8}	3×10^{16}	large molecule	
X-ray	10^{-10}	3×10^{18}	atom	excites electrons in atomic orbitals
gamma ray	10^{-15}	3×10^{23}	atomic nucleus	causes atomic nuclei disintegration

QUANTISATION OF ENERGY

OUTCOMES

- analyse the experimental evidence gathered about black body radiation, including Wien's Law related to Planck's contribution to a changed model of light (ACSPH137)  
 - $\lambda_{\max} = \frac{b}{T}$

BLACKBODY RADIATION

THERMAL RADIATION

Light and heat (thermal radiation) is emitted by hot **solids** and **gases**. Thermal radiation refers to electromagnetic radiation emitted by an object due to its temperature. The characteristics of this radiation depend on:

- Temperature
- The properties of the object

At **lower temperatures**, the wavelengths of the thermal radiation are mainly in the **infrared** region and hence not observable to the eyes. As the temperature of an object begins to increase, it eventually begins to **glow red**. At sufficiently **high temperatures**, it appears to be **white** as in the glow of a hot tungsten filament of a lightbulb.

The light emitted from a source can be analysed using a **spectrometer**. It measures the **intensity** of each **wavelength** of radiation emitted.

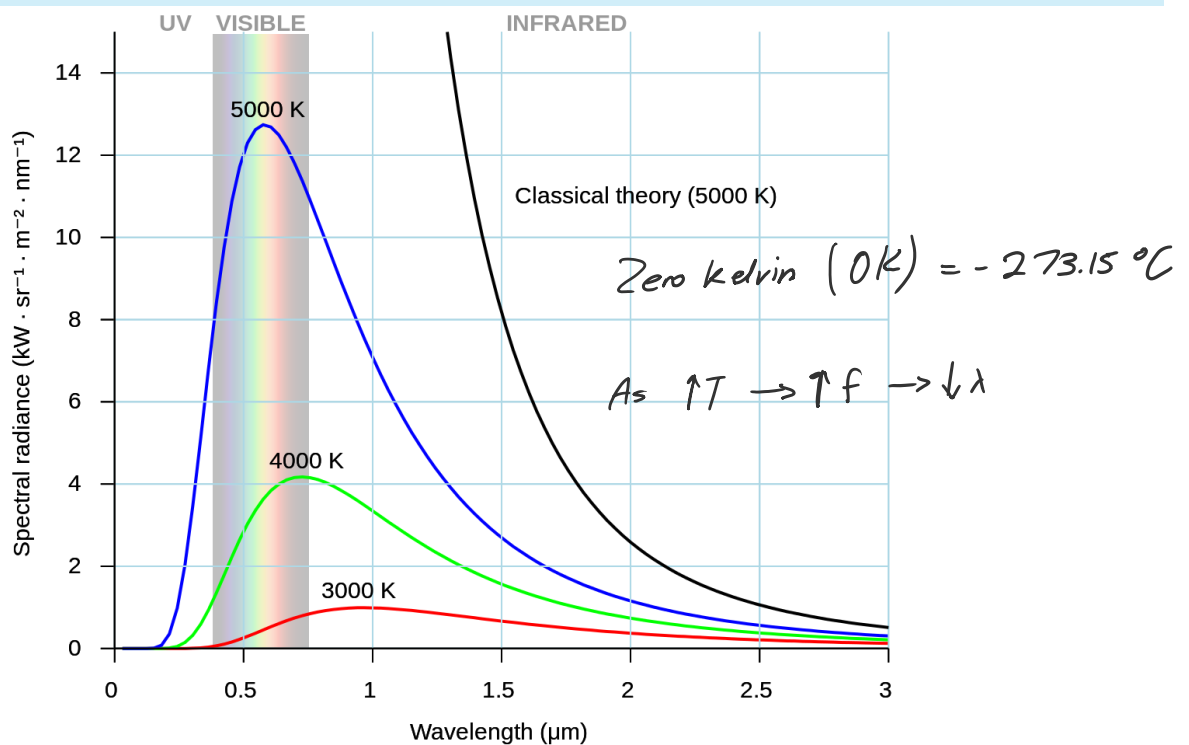
BLACK BODY

The term 'black body' was introduced by Gustav Kirchhoff in 1862. A black body is an **ideal physical object** which absorbs **all electromagnetic radiation incident upon it** (from an external source).

- A blackbody is a **perfect absorber** of electromagnetic radiation over all wavelengths. If the incident radiation is absorbed, **any electromagnetic radiation from the object must be its own** (due to temperature).
- Blackbodies are **perfect emitters** of electromagnetic radiation as it is capable of emitting radiation over all wavelengths.
- The electromagnetic radiation emitted by a blackbody at a constant temperature is called **blackbody radiation**.

The sun is approximately a black body. The light it emits is almost entirely due to its temperature of its surface. The moon, however, is not a black body. The light we observe from the moon is almost entirely a reflection of the sun's. It does not depend on the moon's surface temperature.

NATURE OF BLACKBODY RADIATION



The intensity and wavelength of the electromagnetic radiation emitted by a blackbody is directly related to its **temperature**.

- Blackbodies below around **700 K** produce very little radiation at visible wavelengths and **appear black**.
- Blackbodies **above 700 K** begin to produce radiation at **visible wavelengths**.
- The radiation becomes shorter in wavelengths as the temperature increases.
- At higher temperatures the larger intensity of shorter wavelengths changes the colour of the black body from **red to orange, yellow, white and blue** as temperature increases.

The characteristics for the distribution of intensity vs wavelength in black body radiation are:

- At any given temperature of the black body
 - 1) There is a **peak** in the intensity distribution
 - 2) High frequency radiation are not emitted
- As the temperature of the blackbody increases
 - 1) The intensity at every wavelength increases
 - 2) The total intensity emitted increases
 - 3) The peak of distribution shifts to shorter wavelengths

WEIN'S DISPLACEMENT LAW

Wein's Displacement Law relates the temperature of a blackbody with its peak wavelength.

$$\lambda_{\max} = \frac{b}{T}$$

Where:

- $\lambda_{\max}$: peak wavelength (m)
- b : Wein's displacement constant ($b = 2.898 \times 10^{-3} \text{ m.K}$)
- T : temperature (K)

CLASSICAL EXPLANATION OF BLACK BODY RADIATION

An object at some temperature will emit electromagnetic radiation as the particles in a solid vibrate back and forth and collide with each other exchanging energy. As the particles are vibrating back and forth, they must be **accelerating**, and hence produce electromagnetic radiation.

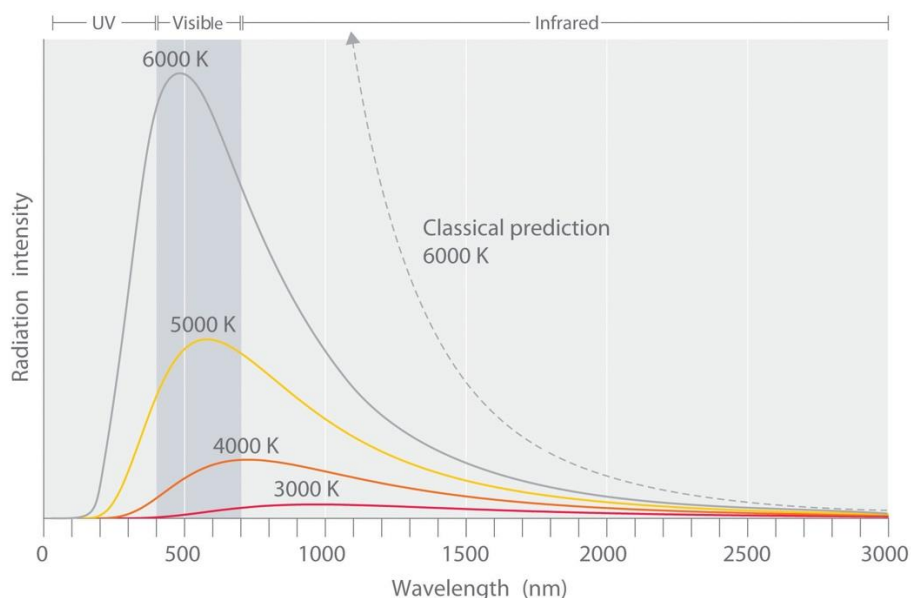
CLASSICAL PREDICTIONS OF THE BLACK BODY SPECTRUM

According to classical theory:

- 1) All frequencies of oscillation are equally likely
- 2) Each particle emits radiation at the same frequency at which it oscillates
- 3) Each particle will emit the same amount of energy on average

Therefore, the **intensity of radiation is higher for high frequencies**, since there are more high frequencies than low frequencies.

The distribution of intensity of radiation over wavelengths is predicted to take the shape as shown below for **any black body at any temperature**,



The prediction (classical curve) does not match the experimental result. It is close however at long wavelengths.

This disagreement was referred to as the **ultraviolet catastrophe** since the disagreement between classical theory and experiment occurred at short wavelengths corresponding to UV.

Feature	Classical Theory	Experimental Result	Conclusion
Behaviour at long wavelengths	Gradual decrease in power at longer wavelengths	Gradual decrease in power at longer wavelengths	Reasonable agreement
Peak Wavelength	Gradual increase in power at short wavelengths. No peak.	Peak wavelength depends on temperature according to Wein's displacement law.	Disagreement
Behaviour at short wavelengths	Sharp increase in power at short wavelengths. Power approaches infinity	Power emitted is 0 at short wavelengths	Disagreement (UV Catastrophe)

Classical theory could not explain the experimental results of black body spectra and a new theory regarding the nature of light was required.

PLANCK'S EXPLANATION OF BLACK BODY RADIATION

Planck's theory was drastically different from classical theory. He said:

- 1) All frequencies of oscillations are **not equally** likely.
- 2) Each particle will **not** emit the same amount of energy on average.

He instead proposed that the intensity of the radiation would increase by **fixed amounts** that depend on the **frequency of the radiation**. These fixed amounts are thought as **discrete packets of energy** or **quanta** (of energy). This became the foundation for modern Physics and quantum theory.

	Hypothesis	Consequence
1 st part of hypothesis	The energy of oscillating atoms that emit electromagnetic radiation is discrete or quantised. We say atoms will occupy discrete energy levels.	If the energy is discrete, the changes in energy are also discrete, going from one level to another When changing energy, the amount of energy that an atom can gain or lose is discrete or quantised.
2 nd part of hypothesis	The frequency of radiation emitted is proportional to the amount of energy released by an atom: $E = hf = \frac{hc}{\lambda}$	The energy released by atoms is in packets or quanta. for an atom to emit high frequencies, it must release a large amount of energy in a single change of energy level.

Planck did not understand the full implications of his theory at first. He only considered this as a mathematical trick that gave him the right equation for the spectrum of a black body, which allowed him to explain Wein's displacement law.

THE ULTRAVIOLET CATASTROPHE

According to Planck's postulates, a discrete packet of radiation or a quantum is emitted or absorbed when an atom undergoes a change in energy level.

Change in energy level → Energy is released or absorbed → Corresponding frequency of radiation

- In order to **emit a short wavelength of radiation** such as UV, or X-ray **atoms need to undergo a large change in energy** that corresponds to the energy of the emitted radiation.
- However, at any given temperature, energy ranges of a certain size do not exist as no single atom has sufficient energy. Atoms thus cannot emit such short wavelengths (high frequencies) radiation at this temperature.
 - o Alternatively, extremely high frequencies of oscillations require extremely high kinetic energies and are therefore extremely unlikely in a black body of any temperature.

PEAK RADIATION (WEIN'S LAW)

According to Planck's postulates, the intensity of radiation emitted of some frequency is proportional to the number of atoms undergoing the change in energy.

- The existence of **peak radiation** is due to certain changes of **quantum states of atoms being more probable at a given temperature**.
- Therefore, the more probable changes of energy results in more intense emission of the corresponding wavelength of radiation.
- The wavelength associated with this change in energy is referred to as the **peak radiation** and its wavelength for a given temperature can be calculated using Wein's displacement law.

At each temperature there is some frequency of oscillation that is most probable for the particles in a black body. Most of the time, there will be more particles oscillating at this frequency than at any other. More quanta of light will therefore be emitted at this frequency (when the particle relax to a lower energy)

EXAMPLE QUESTION 1

Explain how understanding blackbody radiation changed the direction of scientific thinking in the early 20th century.

Classical electromagnetism and thermodynamics would not account for the nature of black body radiation. Physicists using theories in the late 19th century concluded that black body radiation should almost emit light with an intensity that increases infinitely with decreasing wavelength. But in reality, each temperature emits a certain wavelength most intensely – black body spectra have peaks – and extremely short wavelengths should not be emitted at all.

Planck's used the right equation for the black body spectra by imagining the oscillating particles in a black body emit light not in continuous waves (as Maxwell's theory) but in discrete 'quanta', each of the energy $E = hf$. Although Planck took this quantisation to be merely a mathematical trick, Einstein would later develop it into a particle (photon) theory of light.

ELECTRON VOLT (eV)

The **electron volt** is a **unit of energy** defined to be $1\text{ eV} = 1.602 \times 10^{-19}\text{ J}$

The electron volt (eV) is defined as the amount of energy gained (or lost) when a single electron moves across an electric potential difference of 1 volt.

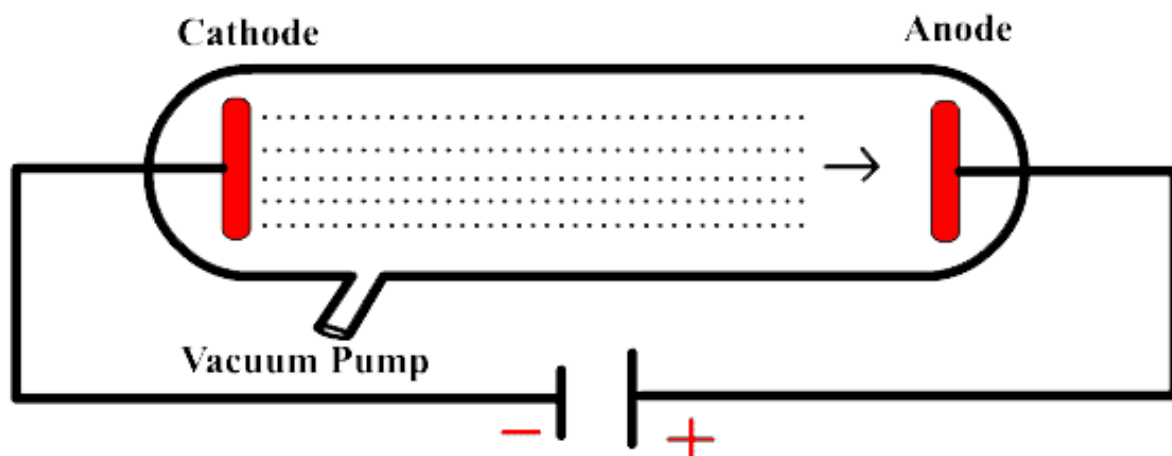
OUTCOMES

- conduct an investigation to examine a variety of spectra produced by discharge tubes, reflected sunlight or incandescent filaments

SPECTRA OF DISCHARGE TUBES

A discharge tube is an evacuated glass tube with **two metal electrodes** inside at **high voltage**. Electrons flow across the tube.

- The **negative** electrode is called the **cathode**
- The **positive** electrode is called the **anode**
- The two electrodes act like **parallel plates** and create an **electric field** between them which **accelerates** the electrons.
- Electrons flow from negative to positive
- The tube must be evacuated (low pressure) to allow the electrons to flow, otherwise the air, which is an insulator, would prevent the flow of electrons.



LIGHT EMISSION FROM DISCHARGE TUBES

Discharge tubes can be filled with low pressure gas of different elements or compounds. Electrons moving through the discharge tube will collide with these gas molecules.

A collision between electrons and gas molecules can result in 3 outcomes:

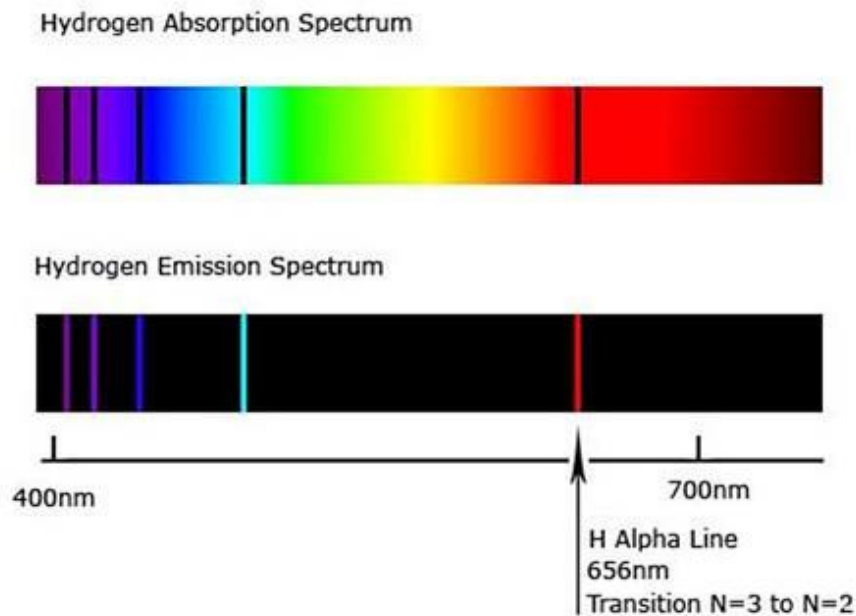
- 1) Low energy collision. The electron is **deflected** by the gas molecule
- 2) High energy collision. Some kind of kinetic energy the electron is absorbed by the gas molecule and then emitted as light. This process is called **fluorescence**.
- 3) Very high energy collision. The electron **ionises** the atom or molecule, resulting in an ion and extra free electron.

Electrons from the cathode ray collide with the gaseous elements. It excites electrons in the gaseous elements and they go up energy levels. When the electron relaxes to a lower energy level, it emits a specific frequency of electromagnetic radiation.

The collisions that result in the emission of light cause the tube to **glow a different colour depending on the gas used**.

A spectrum of light emitted by the discharge tube can be obtained.

- The spectrum shows that **only specific wavelengths are emitted**
- Such spectra are called **line spectra** since they appear as a collection of lines



The emission of discrete wavelengths confirms that the energy states of gas molecules are discrete (quantised).

PHOTON MODEL OF LIGHT (EINSTEIN)

OUTCOMES

- investigate the evidence from photoelectric effect investigations that demonstrated inconsistency with the wave model for light (ACSPH087, ACSPH123, ACSPH137) ⚙️💻
- analyse the photoelectric effect $K_{\max} = hf - \phi$ as it occurs in metallic elements by applying the law of conservation of energy and the photon model of light, (ACSPH119) 🖨️📄

THE PHOTOELECTRIC EFFECT

The **photoelectric effect** is a phenomenon in which **electrons are ejected** from a **metal when light is shone on the metal surface**. Electrons emitted in this manner are called **photoelectrons**. This can occur as metals have a **sea of delocalised electrons** surrounding its cation lattice.

Maxwell's theory of light **could not predict** this, instead predicting that any frequency of light would cause photoelectrons. This led Einstein to develop the **photon model of light**. This is when classical physics and observations **do not match up**.

- Classical theory says that electromagnetic radiation incident on the surface of a metal will cause electrons to oscillate.
- More radiation, of any frequency, will cause the electrons to oscillate more
- Putting in more and more energy can cause an electron to oscillate so violently that it breaks free from the metal. Thus, **any frequency** could cause photoelectron emission, whenever enough energy has been absorbed by the electron.
- This **prediction disagreed with the experiment**

INTRODUCTION AND EXPERIMENTAL SETUP

A device that utilises the photoelectric effect is called a **photocell**.

- An evacuated transparent tube contains a metal plate (photoelectrode) connected to one terminal of a variable voltage source
- Another metal plate is connected to the second terminal of the voltage source
- A voltmeter measures the voltage from the power source
- An ammeter measures any electrons following from the photoelectrode to the other metal plate

The following can be changed:

- The frequency of light
- Intensity of light
- Voltage across cathode and anode

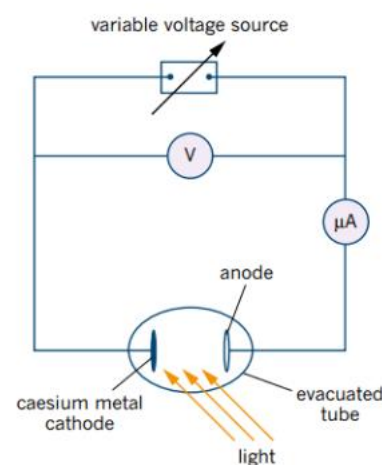


FIGURE 11.2.1 Circuit diagram of an experimental investigation of the photoelectric effect.

Interested in number of photoelectrons per second (**current**) and largest kinetic energy of photoelectrons.

CONCLUSIONS OF THE EXPERIMENT

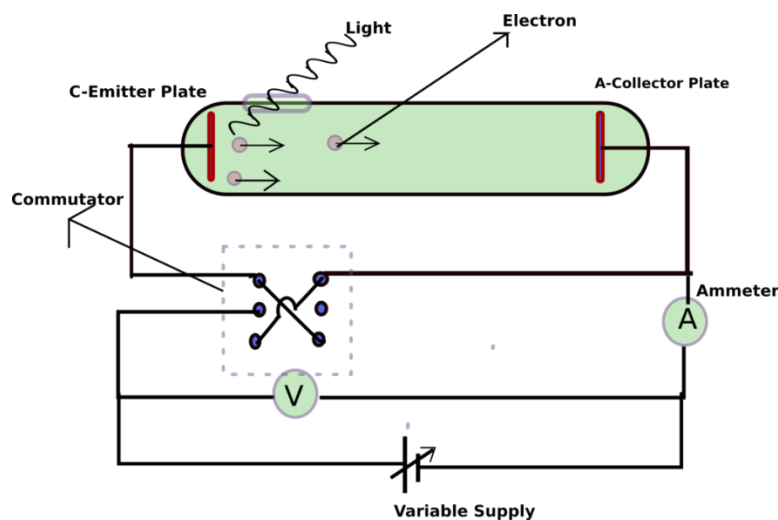
Property	Depends on
Number of photoelectrons per second (current)	Intensity of light
Largest kinetic energy of photoelectrons	Frequency

Note: Maxwell's classical theory predicts that largest kinetic energy of photoelectrons depends also on the intensity of light.

MEASURING MAXIMUM KINETIC ENERGY OF THE PHOTOELECTRONS

To measure the **number of photoelectrons** that reach the terminal per second: measure the current and divide by the charge of an electron.

To measure the **maximum kinetic energy of the photoelectrons**: measure the back voltage at which current is reduced to 0.

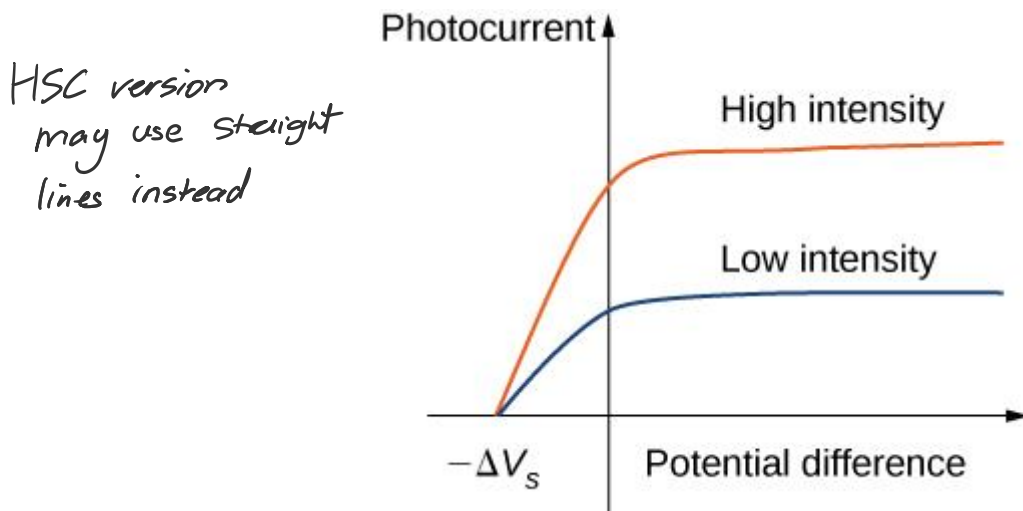


- When A is negatively charged, it is called the **stopping electrode**. Not all the electrons will make it to the anode. Only electrons with enough kinetic energy will reach the anode.
- The work the electric field does on the electron is $W = qV = eV$.
 - o Electrons need $K_{initial} > \Delta K = w = qV$ to reach the anode on the other side when A is negative
- As the stopping electrode is made more **negative**, the electrons will be repelled before reaching it
- At some voltage only those electrons with the **highest kinetic energy** $E_{K(max)}$ will make it. At only slightly higher voltage, no photoelectrons are received at the stopping electrode, indicated by the **current dropping to 0**. This voltage is called the **stopping potential** V_s
- Since the stopping potential was just enough to stop the fastest photoelectrons, the maximum kinetic energy of the photoelectrons is $E_{K(max)} = qV = eV_s$

CHANGING THE LIGHT INTENSITY

- When the evacuated glass tube is kept in the **dark**, the ammeter reads **zero**, indicating that there is **no current** in the circuit.
- When a low intensity of monochromatic (single wavelength) light of **high enough frequency** is shone on plate C, a **current** is detected by the ammeter indicating a **flow of charge** across the gaps between C and A.
- When intensity is increased, the current increases.
 - o 1 photon (with a discrete energy) can be absorbed by 1 electron
 - o This means that electrons undergo a 1:1 interaction with photons to become photoelectrons

CHANGING THE VOLTAGE ACROSS THE PLATE



- 1) Current reaches a limiting value for any particular monochromatic light intensity. The maximum current value corresponds to the case where all photoelectrons are collected at A.
- 2) Increasing the intensity increases the current for any voltage
- 3) At greater negative voltage, more electrons are stopped and repelled before reaching plate A
- 4) The **stopping voltage depends on frequency** but not on intensity. Current is zero beyond the stopping voltage because no photoelectrons are able to overcome the repulsive electric force

EINSTEIN'S PHOTOELECTRIC EQUATION

Einstein was able to explain the photoelectric effect using this equation. The maximum kinetic energy for photoelectrons are:

$$E_{K(max)} = hf - \phi$$

Where:

- $E_{K(max)}$ = maximum kinetic energy of the photoelectrons (J)
- f = frequency of the incident light (Hz)
- ϕ = the "work function", a constant dependent on the metal used

If $hf_0 = \phi$: electron pops out however is recaptured into the metal lattice unless its affected by the electric field.

The minimum amount of work for an electron to be ejected from a given metal is given by ϕ . Hence ϕ determines the threshold frequency (f_0) for electron ejection.

If $hf > \phi$ and $E_{K(max)} > 0 \therefore$ electron escapes by overcoming the work function and flies away with some energy remaining.

If $hf < \phi$ and $E_{K(max)} < 0$ is undefined since the electron is not ejected.

The **work function** represents the minimum energy with which an electron is bound in the metal (usually a few electron volts).

EINSTEIN'S EXPLANATION OF THE PHOTOELECTRIC EFFECT

EINSTEIN'S PHOTON THEORY OF LIGHT

Einstein extended Planck's concept of quantisation to electromagnetic waves in order to explain the observations of the photoelectric effect. He postulated:

- 1) **Light** (or any electromagnetic wave) is a **stream of particles called photons**, each with energy $E = hf$
- 2) Energy of the beam of light is localised to each photon, not spread across the wavefront of the light beam
 - a. Different colours of visible light have different energies.
- 3) Each photon has energy determined by Planck's equation: $E = hf$

Using the photon theory of light in the photoelectric experiment:

- **Frequency** is related to the amount of **energy each photon carries**
- **Intensity** is related to the **number of photons per second**

EINSTEIN'S PHOTON EXPLANATION OF THE PHOTOELECTRIC EFFECT

Einstein's simple view of the photoelectric effect was that a **single photon** gives **all of its energy** to a **single electron** in the metal. The electron uses some of that energy to escape from the metal, and the remainder becomes the kinetic energy of the electron.

$$E = hf \rightarrow E_p + E_K$$

This **one-to-one interaction between photon and electron** occurs because the photon is the smallest unit (quantum) of light and cannot be split up.

- 1) Light is shone on the metal. Each photon of light carries energy $E = hf$
- 2) An electron in the metal absorbs this energy. It requires energy to escape the electrostatic force binding it in the metal. The electrons bound the weakest in the metal require an energy equal to the 'work function' ϕ
- 3) Any left over energy becomes kinetic energy of the electron as it exits the metal. The electron bound with energy ϕ have maximum kinetic energy of $E_{K(max)}$

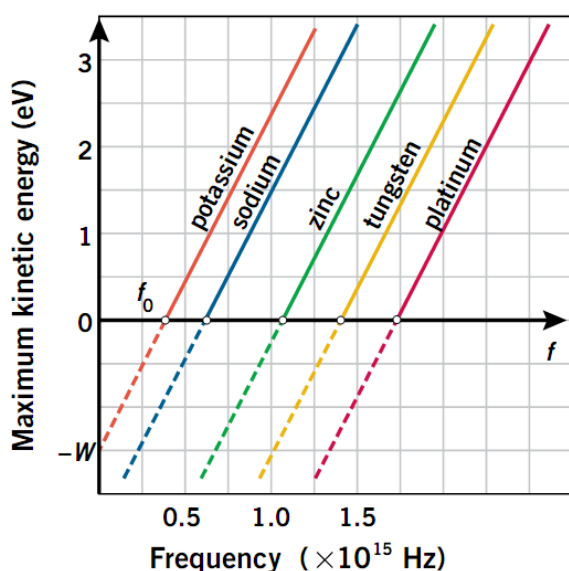
WORK FUNCTION

The work function represents the **minimum energy** with which an electron is bound in the metal (usually a few eV).

- The value of the work function is **different** for **different metals**
- The work function can be determined if we know the threshold frequency (x-intercept)

$$E_{K(max)} = hf - \phi$$

If $E_{K(max)} = 0$ when ($hf = \phi$) then $f = f_0$.



It is in the form of $y = mx + b$.

$$E = hf - \phi$$

Where:

- $m = h$ (Planck's constant. All lines should be parallel to each other)
- $x_{int} = f_0$ (Threshold frequency – minimum frequency of photon to eject electron)
- $y_{int} = -\phi$ (Work function – minimum amount of work to eject an electron)

- All lines are **parallel** since the gradients are equal to Planck's constant 'h'
- The x-intercept (f_0 threshold frequency) is different for different metals, and thus the y-intercept (ϕ work function) is different for metals

SAMPLE QUESTION 1

Discuss how Einstein's contributions changed scientific thinking regarding blackbody radiation.

The gradient of the photoelectric result is Planck's constant, with the relationship $E_{K(max)} = hf - \phi$. The experimental evidence for discrete energy levels in matter (ϕ) and quantised light energy (hf) supports Planck's hypothesis and his explanation of blackbody radiation. Blackbody radiation became more accepted and thinking changed from treating light as a wave to treating light as a particle.

SAMPLE QUESTION 2

Outline Einstein's explanation of the photoelectric effect

Light is a stream of photons (indivisible particles/quanta of light) with energy $E = hf$ per photon. Photons undergo 1:1 interaction with electrons in a metal, wherein the electron absorbs the photon in its entirety. Metals have a work function ϕ , a minimum amount of work to escape the metal. If incident photons have energy $hf < \phi$, electrons are not ejected. If $hf > \phi$ electrons are ejected with kinetic energy: $E_{K(max)} = hf - \phi$. The number of ejected electrons for $f \geq f_0$ corresponds to the number of photons, hence photocurrent is directly proportional to intensity.

EFFECTS OF CHANGING FREQUENCY IN THE PHOTOELECTRIC EFFECT

Classical theory prediction:

- Below a certain intensity, where the light has little or no effect on the metal, changing the frequency will not produce a photoelectron
- Electrons should be emitted at any frequency, given the intensity is high enough

Experimental observation:

- No electrons are emitted if the incident light frequency falls below some threshold frequency, f_0 , which is characteristic of the material being illuminated

Einstein's explanation:

- If the energy of the incoming photon is not equal to or greater than the work function, ϕ , the electrons will never be ejected from the surface, regardless of the intensity of the light

EFFECTS OF CHANGING INTENSITY IN THE PHOTOELECTRIC EFFECT

Classical theory prediction:

- Increasing intensity increases the rate at which electrons were ejected from the metal surface and so the photocurrent would increase
- The ejected electrons will also have greater kinetic energy
- If the intensity were reduced dramatically, it would take a long time for electrons to be ejected, but eventually this will occur

Experimental observation:

- If the light frequency exceeds the threshold frequency, a photoelectric effect is observed, and the number of photoelectrons emitted is proportional to **light intensity**
- The maximum kinetic energy of the photoelectron is **independent** of light intensity

Einstein's explanation:

- If the light intensity is doubled, the number of photons is doubled, which doubles the number of photoelectrons emitted. However, their kinetic energy depends only on the light frequency and the work function, not on the light intensity.

EFFECTS ON KINETIC ENERGY OF PHOTON ELECTRONS

Classical theory prediction:

- As the light intensity increases, kinetic energy of the photoelectrons increases

Experimental observation:

- The maximum kinetic energy of the photoelectron increases with increasing light frequency
- The maximum kinetic energy of the photoelectron is independent of light intensity

Einstein's explanation:

- The energy absorbed by the weakest bound electron is $E = hf$. This is used to overcome the work function and the remainder becomes kinetic energy. Greater photon energy means a greater $E_{K(max)}$. An electron can only absorb 1 photon at a time

EFFECTS ON EMISSION TIME OF PHOTOELECTRONS

Classical theory prediction:

- The electrons require some time to absorb the incident radiation before they acquire enough kinetic energy to escape from the metal
- For low intensity light, the time for the electrons to be emitted would be long, in the order of minutes

Experimental observation:

- Electrons are emitted from the surface almost instantaneously if the frequency of incident light is $> f_0$ even at low light intensities.
- Electrons don't require a gradual build-up of energy

Einstein's explanation:

- There is one to one interaction between photons and electrons. When an electron absorbs one photon of energy, it is instantly ejected.

DUALITY OF LIGHT

Quantum mechanics describes light as having both wave and particle characteristics depending on the situation. Hence it has dual wave-particle nature.

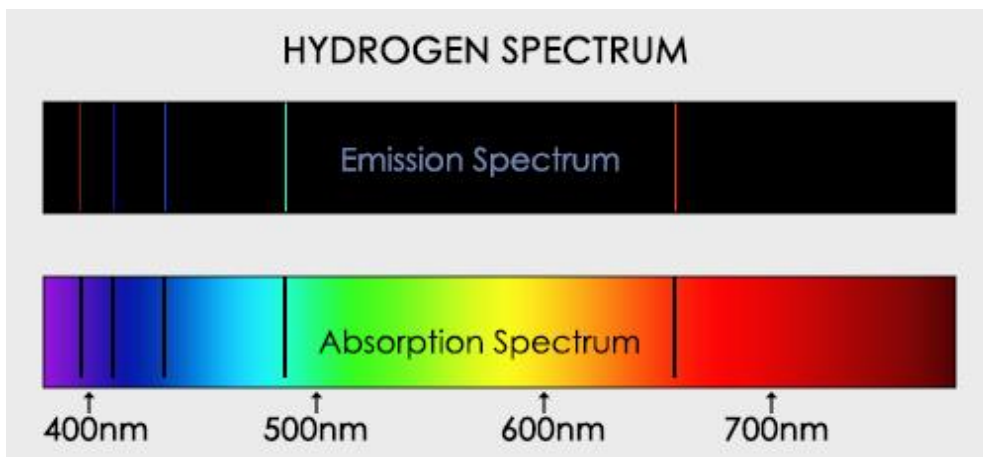
SPECTROSCOPY

OUTCOMES

- investigate how spectroscopy can be used to provide information about:
 - the identification of elements
- investigate how the spectra of stars can provide information on:
 - surface temperature
 - rotational and translational velocity
 - density
 - chemical composition

ATOMIC SPECTRA

Atomic spectra refer to the **spectrum of light absorbed or emitted by specific elements**. There are two types of atomic spectra, absorption and emission.

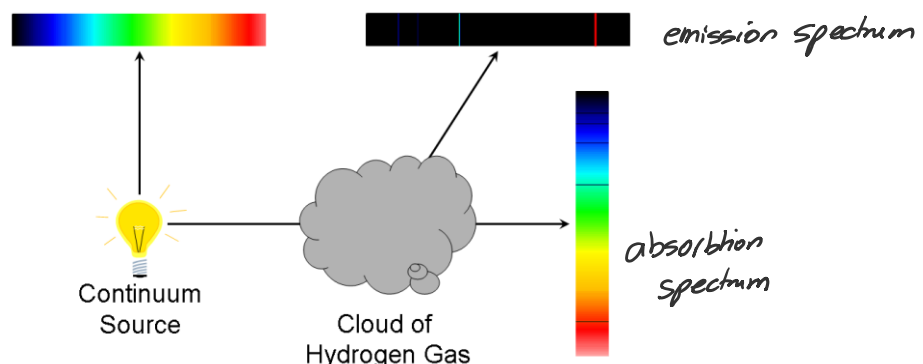


The wavelength at which light is absorbed and at which light is emitted are **exactly the same** for a given element.

ABSORPTION SPECTRA

Absorption spectra involves passing light with a wide wavelength range (white light) through a sample, and recording the spectrum of light that is transmitted.

- The wavelengths **not absorbed** will pass through and **appear** in the spectrum.
- The wavelengths that **are absorbed** will be **missing** in the spectrum and will appear as **dark lines**.



EMISSION SPECTRA

Emission spectra are the spectra of light emitted by a sample after the sample has been given energy in some form.

- A sample will absorb energy (e.g. from heat or from passing a current through it in a discharge tube)
- The sample will release the energy by emitting light. Only **specific** wavelengths of light are emitted.

EXPLAINING ATOMIC SPECTRA USING BOHR'S MODEL

In 1913, Niels Bohr developed a model of the atom that explained the interactions between electrons in atoms and light.

- Bohr postulated that electrons within an atom existed in **specific, quantised energy levels**. This is based on Planck's theory of quantisation of energy.
- He also postulated that the only way for an electron to move between these levels was via the emission or absorption of a photon. This relies on Einstein's photon theory light.
- The energy of the photon absorbed or emitted is identical to the energy difference between the electron energy levels.

EMISSION SPECTRA

An emission spectrum is produced when **excited** electrons in an atom **return to a lower energy level** and emit photons of characteristic wavelength.

An element in an excited state possess energy E_{upper} , greater than when its at a lower state E_{lower} .

- When the element returns, or relaxes to a lower energy state, the excess energy must be released. The excess energy is equal to the difference between the two energy levels ΔE
- That energy is released when a single photon with energy ΔE , and hence at a specific wavelength.
- The wavelength of an emitted photon is detected as a thin line of colour

$$\Delta E = E_{upper} - E_{lower} = hf = \frac{hc}{\lambda}$$

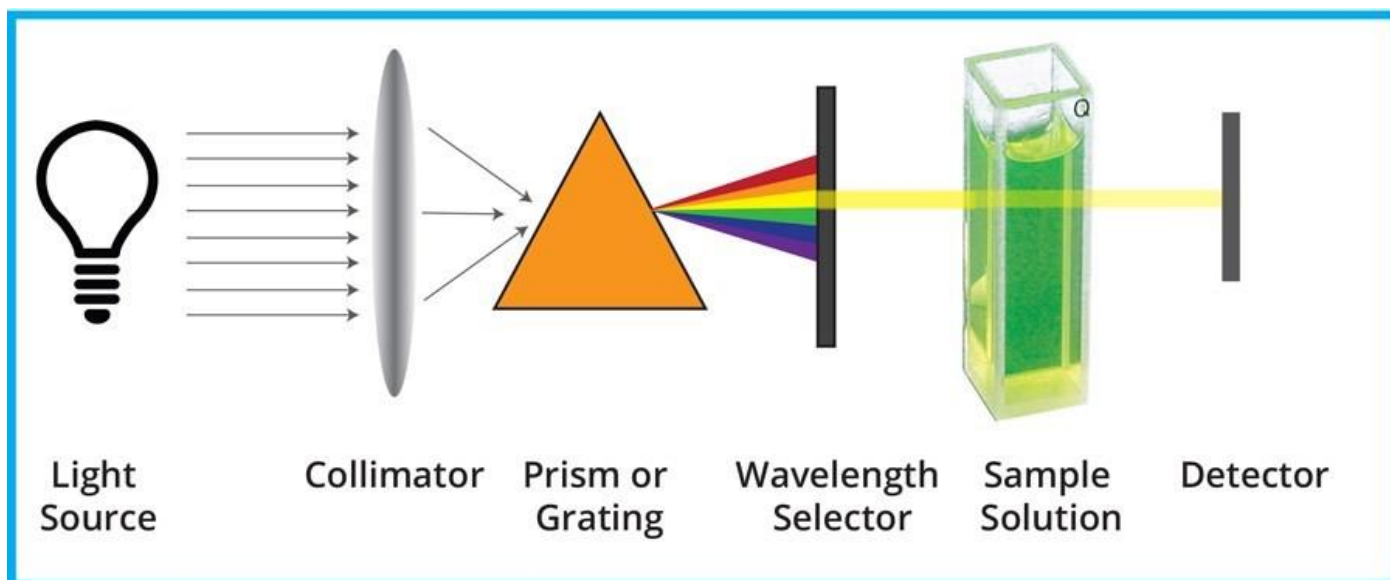
ABSORPTION SPECTRA

An absorption spectrum is produced when electrons in atoms absorb photons of certain wavelengths and become excited from lower to higher energy levels.

- An absorption only occurs when the photon's energy **matches** the difference in energy, ΔE , between two energy levels
- The intensity of the photons passing through a sample containing the element is attenuated (i.e. decreased) because of absorption.
- The absorption spectrum will appear as **black lines** against a continuous spectrum (containing all wavelengths of visible light).

ABSORPTION SPECTRUM

- Light from a source (a lamp) is passed through a prism or grating to disperse it (separate it out into different wavelengths)
- A monochromator slit is used to select one wavelength.
- This wavelength is passed through a sample and a detector measures how much light passes through. This is compared to how much light passes without the sample to calculate the fraction absorbed.
- The wavelength is changed and the measurements is repeated for different wavelengths.

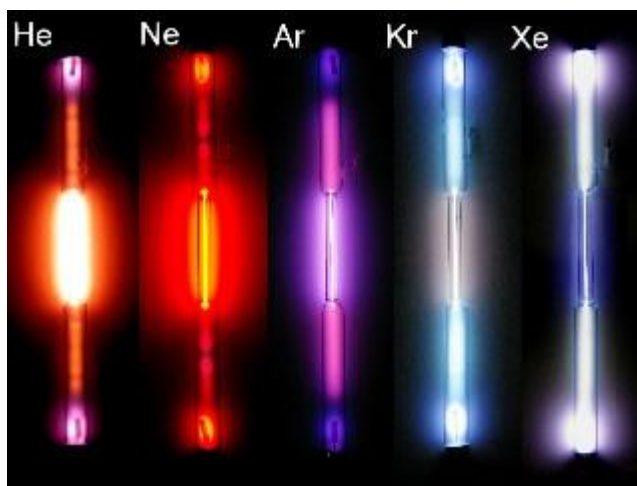


Part	Function
Lamp	Produces a wide, continuous spectrum (e.g. black body spectrum)
Slit	Creates a thin beam of light
Prism	Disperses the light to form. The rainbow can be moved by rotating the prism
Monochromator slit	Positioned so as for a specific wavelength to pass through it. Select a different wavelength by moving the slit, or rotating the prism.
Sample	Sample to be tested. Will absorb some wavelengths of light.
Detector	Measures the intensity of light

SPECTROSCOPY OF DISCHARGE TUBES

Discharge tubes can emit light via excitation of the gas particles within the tube. The atoms **absorb energy** via collision with the electrons and then **re-emit energy at specific wavelengths** – this process of absorption and re-emission is known as **fluorescence**.

- This means that the emitted light of a discharge tube depends on the emission spectra of the gases contained within.



Discharge tubes can be further coated to absorb this light produced by the gas in the tube and re-emit it as different colours. This is how **fluorescent lights** operate.

- The coatings are called phosphors
- Using different phosphor coatings, approximations can be made to white light, warm sunlight and a variety of other spectra.

IDENTIFICATION OF ELEMENTS USING SPECTROSCOPY

All elements have characteristic absorption and emission spectra. This uniqueness of each element's atomic spectrum means that atomic spectra can be used to **identify** the presence of particular elements. The use of spectra to identify the contents of sample is **spectroscopy**.

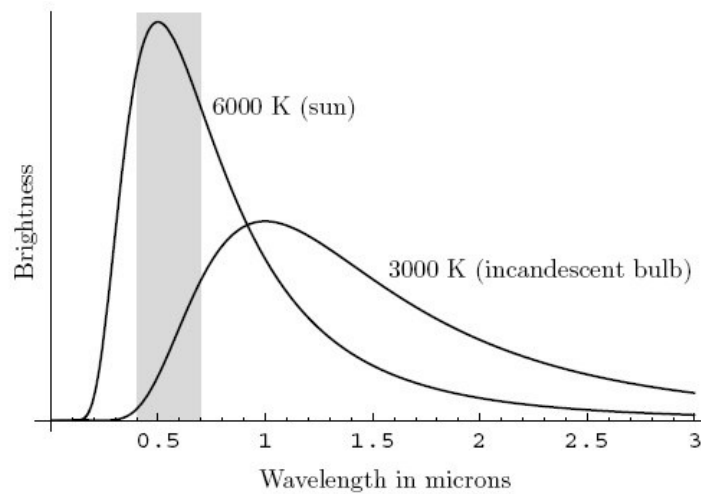
SPECTROSCOPY OF INCANDESCENT FILAMENTS

An incandescent light bulb functions by heating a metal filament, which then emits light and heat. These bulbs emit light via **black body radiation**, not by excitation of particular elements. This means that the light spectrum is

- **Smooth and continuous**
- Covering a **broad** range of wavelengths
- Dependent on the temperature of the filament

Incandescent filaments **require high temperatures** in order to emit visible light.

- Most lightbulb filaments operate at temperatures 2000 – 3000 K
- In order to withstand this temperature, the filament is made mostly out of tungsten which has a high melting point.
 - o 10% efficiency. Incandescent bulbs emit more infrared than visible light.



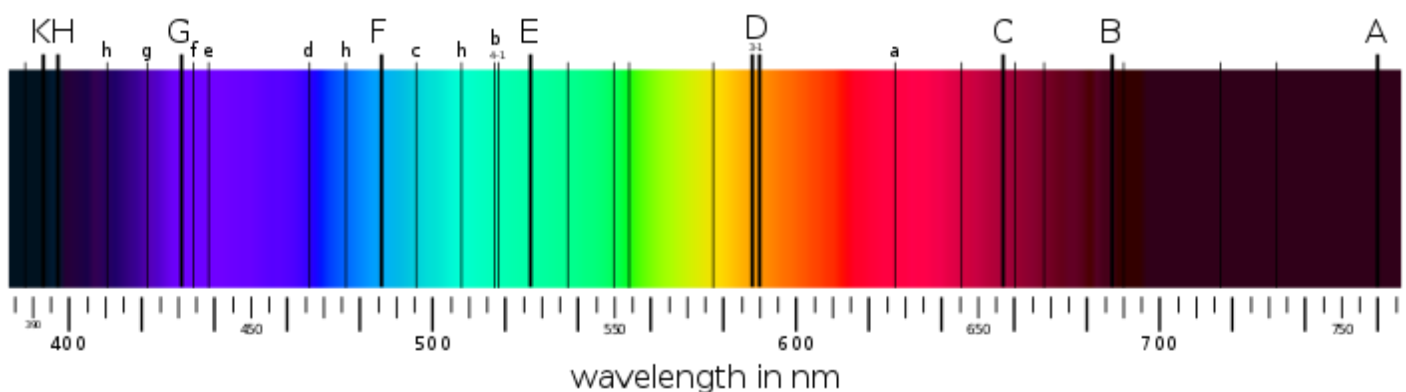
SPECTROSCOPY OF REFLECTED SUNLIGHT

By measuring the spectra of **reflected sunlight** from an object in space and comparing it to known spectrum of the Sun reveals the **absorption spectrum** of that object.

- Whatever light is missing from the reflection **must have been absorbed** by the object.
- If the absorption spectra of certain elements or compounds are known, they can be identified.

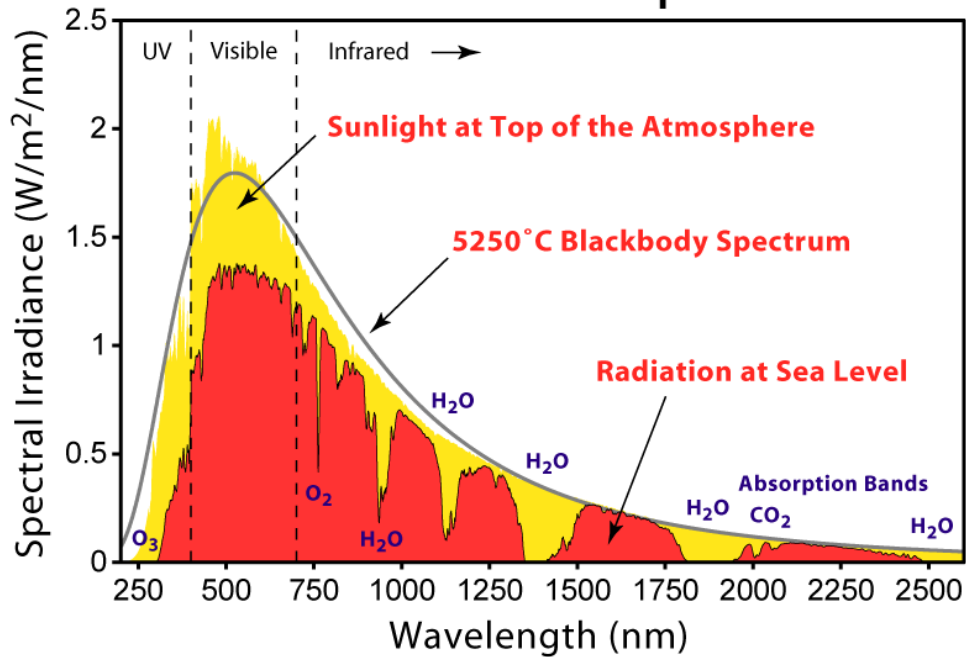
This absorption can be seen in the Earth's atmosphere.

- The light that reaches Earth's surface has specific wavelength ranges missing due to the presence of H_2O , O_2 and CO_2 in the atmosphere which absorb those wavelengths



Fraunhofer lines are a set of spectra lines. The lines were originally observed as **dark features** in the **absorption spectrum**.

Solar Radiation Spectrum



STELLAR SPECTROMETRY

Astronomy is the study of celestial objects such as stars, planets, comets and galaxies.

SURFACE TEMPERATURE

Stars are excellent approximations of **blackbody radiators**. Consequently, the **peak wavelength** of a stellar spectrum allows the surface temperature of a star to be accurately determined.

$$\lambda_{\max} = \frac{b}{T}$$

CHEMICAL COMPOSITION OF STARS

Any star can be considered to consist of a **hot core** surrounded by a **cooler atmosphere**.

- Various ions, atoms and molecules in the cooler atmosphere absorb characteristic wavelengths of light as they pass through it.
- This results in an **absorption spectrum** whose underlying continuum is that of a blackbody.

The chemical composition of **stellar atmosphere** is determined by **comparing** the observed absorption lines with **known** spectra to identify the atoms and molecules present.

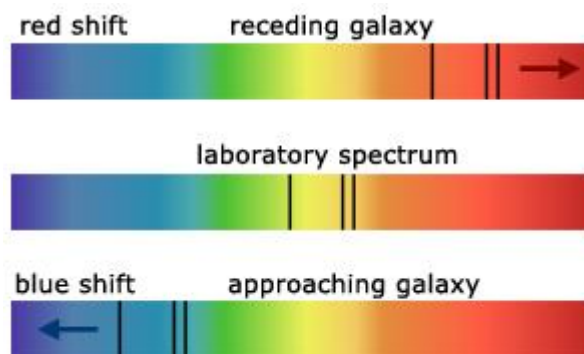
The absence or presence of certain **spectral features**, and their strengths, are used to **classify stars**. Since what spectral lines appear also depend strongly on temperature, the current stellar classification system exhibits strong correlation with colour.

TRANSLATIONAL VELOCITY – DOPPLER SHIFT

The rotational and translation velocity of a star can be determined by examining the results of the **Doppler Effect** on stellar spectra.

- The doppler effect refers to an apparent change in frequency of a wave due to **relative motion** between the source and the observer
- This effect applies to **light** – a frequency shift will occur if the source (star) and the observer (Earth) are moving with respect to each other
- The observed frequency of light f' will differ from the emitted frequency f depending on whether the star and Earth move towards or away from each other and their relative speed.

$$f' = f \frac{v_{\text{wave}} + v_{\text{observer}}}{v_{\text{wave}} - v_{\text{source}}}$$

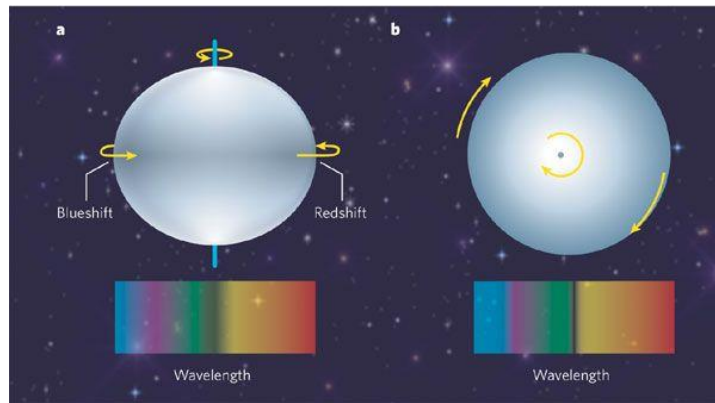


- If an object is moving **away** from the Earth the spectral lines would appear at lower frequency (longer wavelength = redder). Thus those spectral lines would appear to be **shifted** towards the **red end** of the spectrum. This is called **redshift**.
- If the object is moving **towards** the Earth the spectral lines would appear at higher frequencies (shorter wavelengths = bluer). Thus those spectral lines will appear to be **shifted** towards the **blue end** of the spectrum. This is called **blueshift**.

ROTATIONAL VELOCITY – DOPPLER BROADENING

Rotational broadening occurs for any **rotating star**.

- As the star rotates, one side of the star will be moving away from the Earth and the opposite side will be moving towards Earth.
- The edge (limb) of the star that is **moving away from the Earth** causes **red-shifting** of spectral lines
- The centre of the star is moving perpendicularly to the line of sight, so causes no doppler shifting of spectral lines
- All points between the centre and edges of the star causes varying degrees of red-shift/blue-shift. The magnitude of the shifting increases towards the edges.



The degree of broadening in the star's spectrum reveals the rate of rotation. The faster a star rotates, the wider the absorption bands will appear in the spectrum.

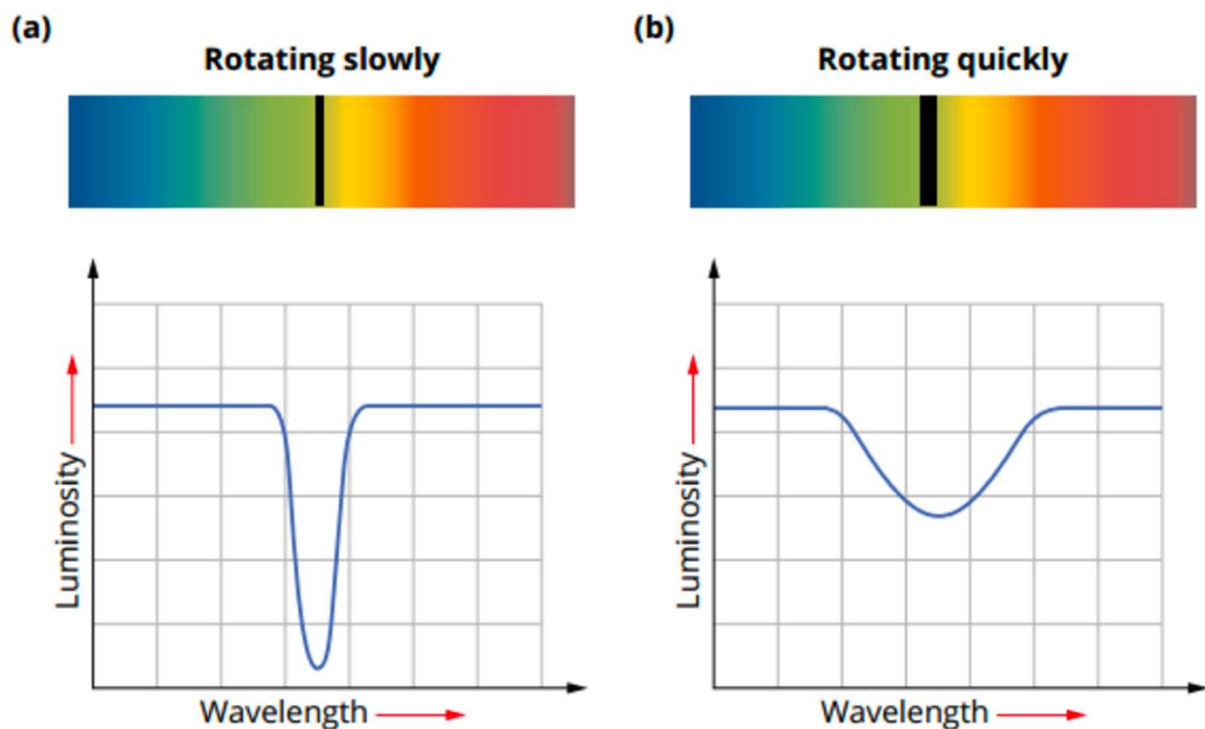
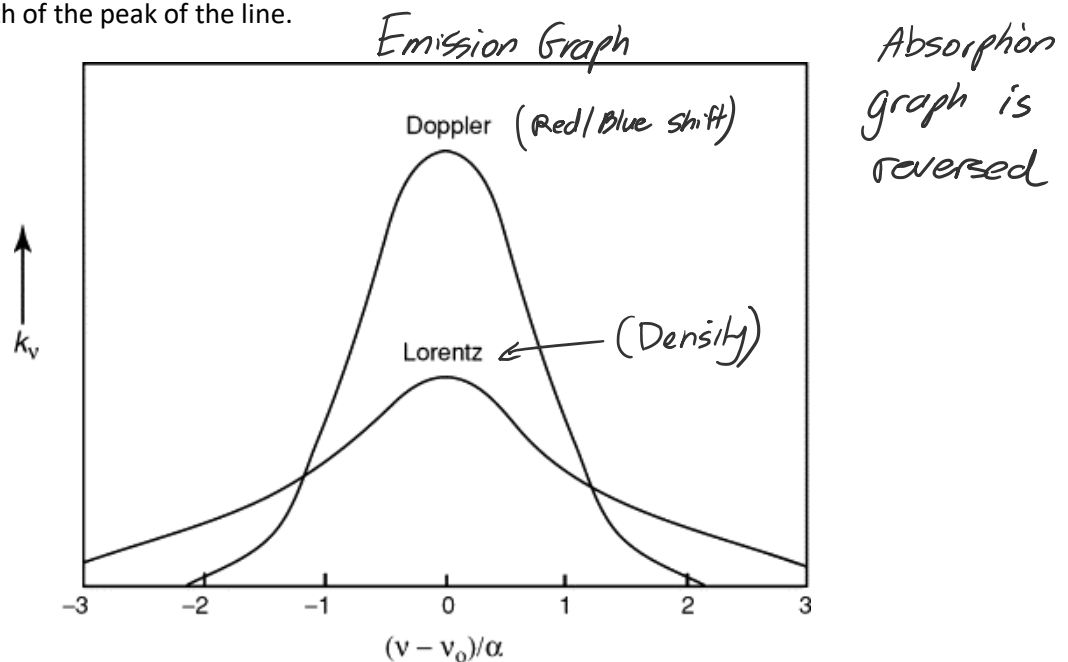


FIGURE 9.2.13 A star's rotation influences the width of its absorption bands. The faster the star spins, the wider the bands.

Pressure broadening is due to collisions between particles, and results in the **broadening of the “wings”** of a spectral line, not the whole line. The wings of a spectral line are the parts near the continuum. Thus, pressure broadening does not affect the width of the peak of the line.



The pressure of a stellar atmosphere is primarily determined by its **temperature** and **volume**. A larger volume results in lower pressure gas, for a given temperature.

The **density** of a gas is directly related to **pressure**. Density determines the **rate of collisions** between particles.

- Lower pressure means lower density and lower collision rate
- Collisions between particles transfer momentum and energy. This allows the particles with highest/lowest kinetic energy to gain/lose a bit more energy. This increases the upper and lower limits of the distribution of random particle velocities. Hence, wings are broadened but peaks are not.

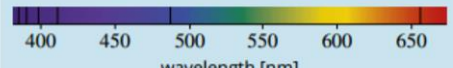
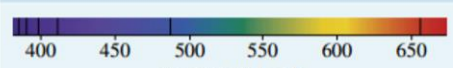
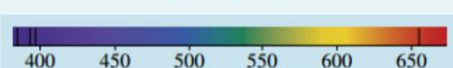
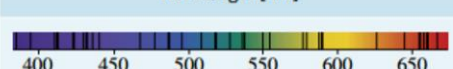
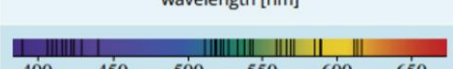
At low pressure:

- Low pressure occurs at large volumes, such as **large stars**
- The **density** is so low the collision rate is low; hence pressure broadening is minimal and lines remain relatively narrow.

At high pressure:

- High pressure occurs at small volumes, such as **compact stars**
- The density is high, so the collision rate is high, hence pressure broadening is significant, and the wings of lines are broadened.

TABLE 9.2.2 The OBAFGKM classification of the spectra of stars.

Spectral type	Approximate temperature (K)	Main characteristics	Spectrum	Colour
O	50000–28000	relatively few lines; the lines of ionised helium	 400 450 500 550 600 650 wavelength [nm]	blue
B	28000–10000	the lines of neutral helium	 400 450 500 550 600 650 wavelength [nm]	blue–white
A	10000–7500	very strong hydrogen lines	 400 450 500 550 600 650 wavelength [nm]	white
F	7500–6000	strong hydrogen lines; ionised calcium lines; numerous metal lines	 400 450 500 550 600 650 wavelength [nm]	yellow–white
G	6000–4900	strong ionised calcium lines; numerous strong lines of ionised and neutral iron and other metals	 400 450 500 550 600 650 wavelength [nm]	yellow
K	4900–3500	strong lines of neutral metals	 400 450 500 550 600 650 wavelength [nm]	orange
M	3500–2000	titanium oxide streaks	 400 450 500 550 600 650 wavelength [nm]	red

SPECIAL RELATIVITY

OUTCOMES

- analyse and evaluate the evidence confirming or denying Einstein's two postulates:
 - the speed of light in a vacuum is an absolute constant
 - all inertial frames of reference are equivalent (ACSPH131)
- investigate the evidence, from Einstein's thought experiments and subsequent experimental validation, for time dilation $t = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$ and length contraction $l = l_0 \sqrt{1 - \frac{v^2}{c^2}}$, and analyse quantitatively situations in which these are observed, for example:
 - observations of cosmic-origin muons at the Earth's surface 🖥️ 📺
 - atomic clocks (Hafele–Keating experiment) ⚙️ 🖥️ 📺
 - evidence from particle accelerators ⚙️ 🖥️ 📺
 - evidence from cosmological studies 🖥️

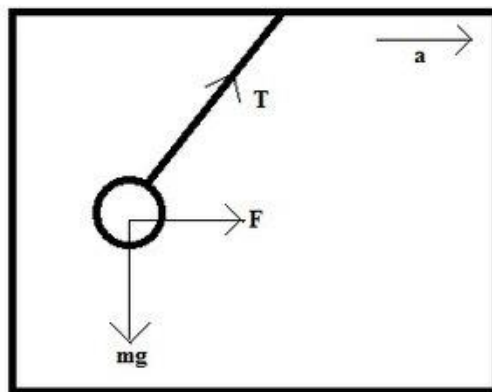
FRAMES OF REFERENCE

A **frame of reference** is a set of coordinate axes relative to which position and motion can be measured. There are 2 types of frame of references: **inertial** and **non-inertial**.

- A **inertial** frame of reference is a **non-accelerating** frame of reference (can be in constant motion).
- A **non-inertial** frame of reference is an **accelerating frame of reference** (net force is non-zero)

DISTINGUISHING BETWEEN FRAME OF REFERENCES

A pendulum can be used as an **accelerometer**. It can be used to deduce the acceleration of a frame from the direction in which the pendulum hangs.



The acceleration of the frame appears to the **observer inside the frame** as a **fictitious force** to the **opposite direction of acceleration**.

- Gravitational force and this fictitious force vector add to give a net force in the same direction that the pendulum is hanging.
- The direction of the **apparent net force** results in acceleration which determines the motion of projectiles **inside** the non-inertial frame. The **fictitious force** does not obey Newton's 3rd Law of motion.

GALILEO'S PRINCIPLE OF RELATIVITY

The Galilean Principle of Relativity proposed in 1632 states “*The mechanical laws of physics are the same for every observer moving uniformly with constant speed in a straight line*” (i.e. for every observer in an **inertial frame of reference**).

The principle of relativity states that there is no **physical way to differentiate between a body moving at constant speed and at rest**. It can be concluded that the principle of relativity holds true for any frame of reference that is not accelerating.

- The laws of physics will not be able to tell you whether you are moving or not. Any experiment will give the same result as if at rest.
- This means that **motion at constant velocity cannot be detected** unless there is a second frame of reference to compare it to.
- Any detected motion of an object is thus **relative** to some frame.

SPECIAL THEORY OF RELATIVITY

THE SPEED OF LIGHT

In the 19th century, scientists began to consider the properties that a **medium** for the propagation of light would require. They called this hypothetical medium the **luminiferous aether**.

- It was proposed that aether formed an **absolute frame of reference** in which the universe and light waves have a **fixed velocity relative to the aether**.
- Hence, light would travel at $3 \times 10^8 \text{ m/s}$ relative to the aether, in the aether reference frame.

This aether theory had certain problems:

- There was no experimental evidence that supported that the aether existed.
- The Earth's motion around the Sun would imply that the Earth moved relative to the aether. This would require the speed of light relative to the Earth to change, which was not detected.

Einstein found a solution to these problems in his **Special Theory of Relativity**, in which he proposed that light does **not require** a medium to propagate.

EINSTEIN'S FIRST THOUGHT EXPERIMENT

“Suppose I am sitting in a train travelling at the speed of light. If I hold a mirror in front of me, will I see my reflection?”

- Answer: No
 - o If the train is travelling at the speed of light, light from his face would not reach the mirror in order to be reflected back.
 - o Since he cannot see his own reflection in the mirror, he would know that the train was travelling at the speed of light **without having to refer to an external point** (violates the Galilean Principle of Relativity), as the observer is able to detect their motion despite being in an inertial frame of reference.
- Answer: Yes

- This means that the light would travel at its normal speed relative to the train. This does not violate the principle of relativity.
- However, based on the addition of velocities, it means that the light inside the train should be travelling at twice the speed of light relative to a stationary observer outside the train.
- Einstein concluded that the **principle of relativity can never be violated**, therefore the true answer is yes, he would see his reflection.
 - He resolved the issue of the speed of light for the two observers by stating that the **speed of light is constant** for any observer. Therefore, the **stationary observer** and the **moving observer** must **perceive distance and time differently** (spacetime). He upheld this as a basic law of physics.
 - Einstein believed in the uniformity of physics. (That a law should be true for all cases, no matter what)

PRINCIPLES OF EINSTEIN'S SPECIAL THEORY OF RELATIVITY

Einstein's Special Theory of Relativity states that:

- 1) The laws of Physics are the same in all inertial frames of reference. (Galilean principle of relativity).
- 2) The speed of light in vacuum has the same value for all observers, regardless of their state of motion or the state of motion of the source. (Only applies in inertial frames).

The speed of light in vacuum is the same value $c = 3 \times 10^8 \text{ m/s}$ in all inertial frames of reference.

This was based on the principle that there was no motion of absolute motion (hence no absolute reference frame), combined with the experimental evidence that the speed of light was always the same, and light appeared not to need aether to travel through.

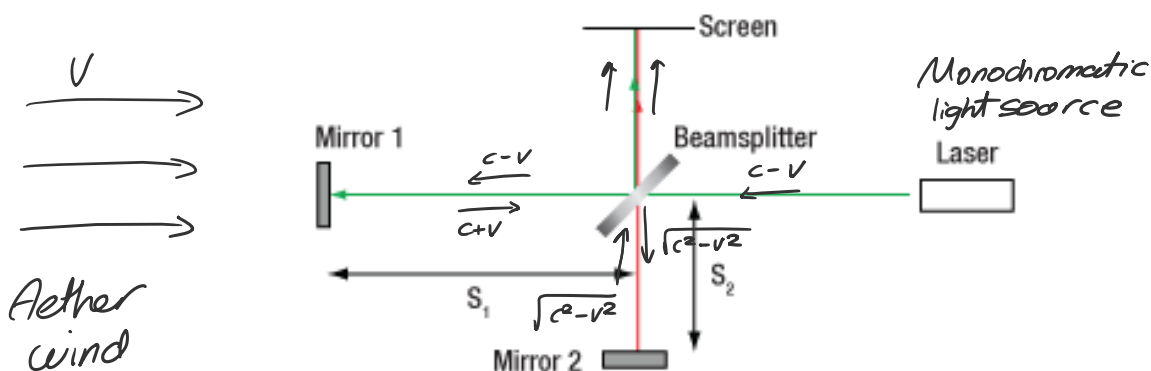
EVIDENCE OF A CONSTANT SPEED OF LIGHT

MICHELSON-MORLEY EXPERIMENT

The aim of the experiment was to **measure differences in the speed of light relative to Earth due to Earth's motion relative to the aether**, called aether wind (which [aether] acted as an absolute frame of reference)

- The results of the experiment **failed to measure any differences** in the speed of light. It was able to support Einstein's theory that the speed of light was constant in all inertial frames of reference.

In their experiment, they directed light at an angle of 45 degrees to half-silvered, half transparent mirror, known as a beam splitter. It splits the beam by allowing half of it through and reflecting the other half.



The experimental tool used is the Michelson interferometer:

- Suppose one of the arms of the interferometer is aligned along the direction of motion of the Earth through space (i.e. through the aether)
- The motion of the Earth is the opposite direction to the aether wind.

The experiment was set up such that the beams interfered. When the two beams are reflected from mirror 1 and 2, they would recombine and interfere.

The interferometer is **rotated by 90 degrees** and the **interference pattern is monitored during the rotation**.

- The rotation would cause the speed of light in each arm to change due to the direction of velocity with respect to the aether wind.
- The speed of light changing would mean the two beams arrived at the detector at different times to before. Hence, their relative phase would be different.
- This would cause the **interference pattern to change during the rotation**.

However, the result of the Michelson-Morley experiment was that **no change in the interference pattern was measured**. This meant that the speed of light in each arm was not affected by the orientation or motion of the Earth.

CONSEQUENCES OF SPECIAL RELATIVITY

In classical physics **space and time are absolute** (same for all observers) and **motion is defined by them**.

In Einstein's Special Theory of Relativity, it is the **speed of light that is absolute** and **space and time** (spacetime) **change to accommodate this**.

- Time and space must change in order to **keep the speed of any beam of light same in all inertial frames of reference**.
- The length of an object and time taken to do something depends on the motion of the observer. Different observers measure **different lengths for the same object**, or a **different time taken to do something**.

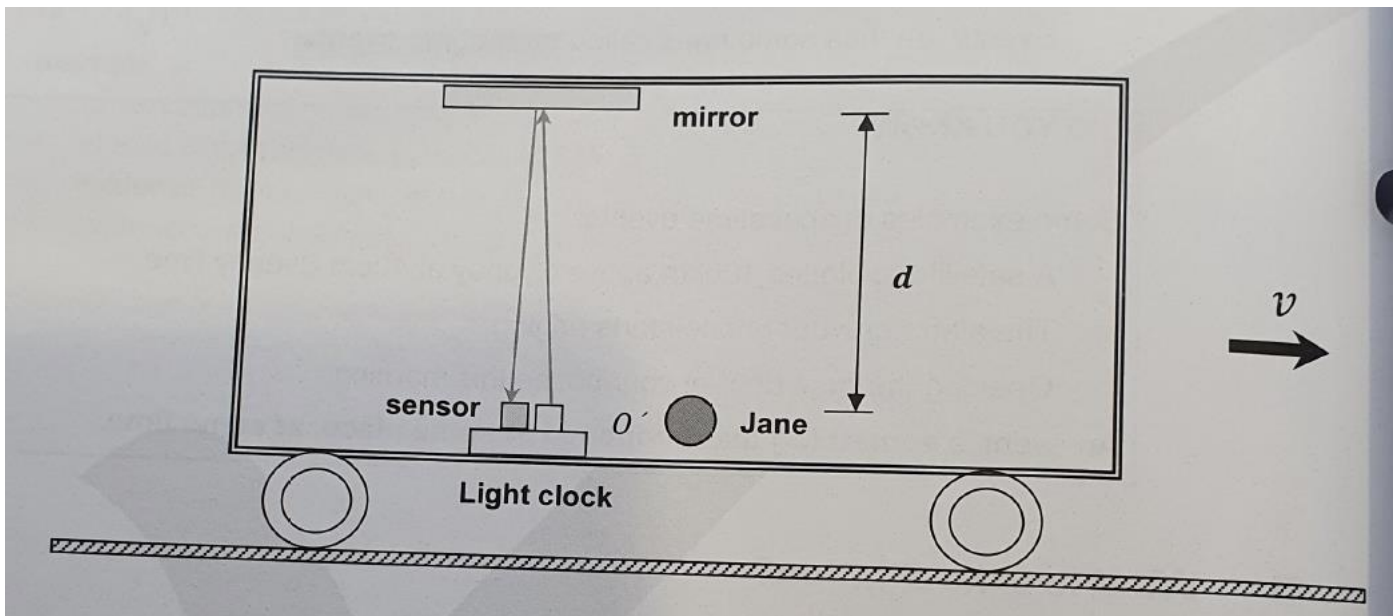
TIME DILATION

According to a stationary observer, a moving clock runs slower than an identical stationary clock. This effect is known as time dilation. Observers in different inertial frames will measure different time intervals between a pair of events.

TIME DILATION THOUGHT EXPERIMENT

A light beam seen by an observer **on** a train

- Consider a train moving to the right with the speed ' v '
 - o A mirror is fixed to the ceiling of the vehicle and Jane at O at rest in this system holds a laser at a distance ' d ' below the mirror.
 - o At some instant the laser emits a pulse of light directed towards the mirror, and at some time after reflecting from the mirror, the pulse arrives back at the laser.



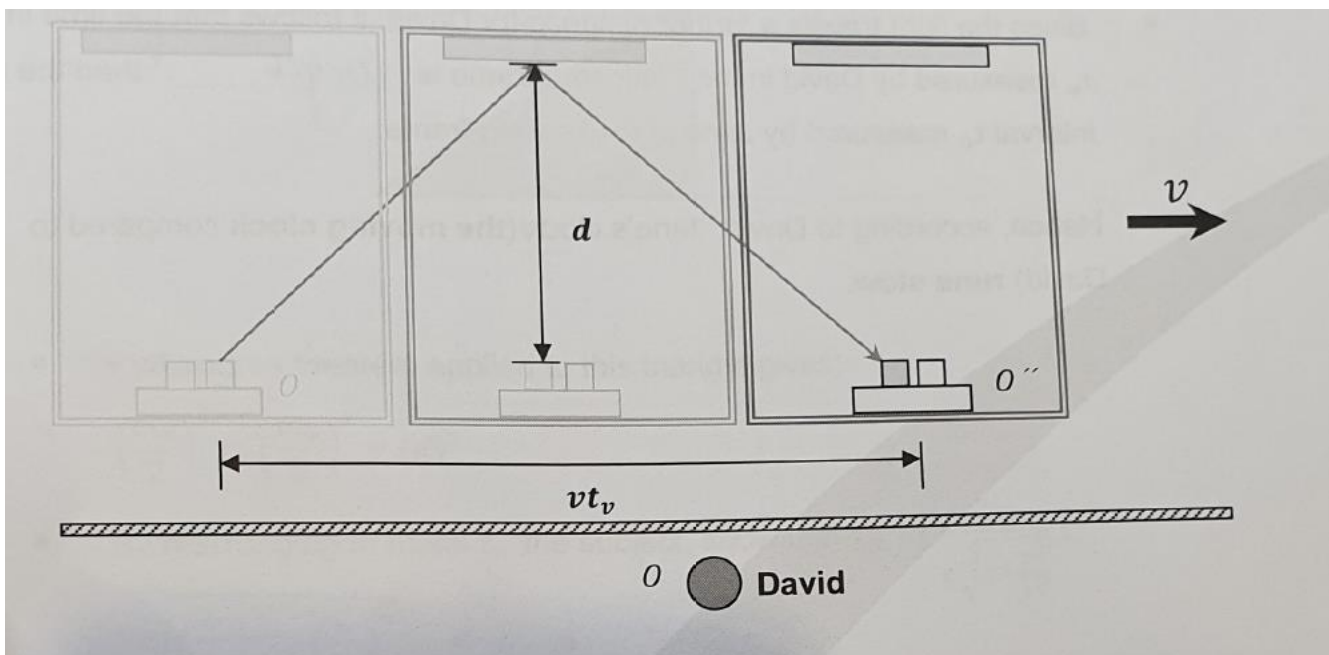
- Jane carries a clock which she uses to measure the time interval ' t_0 ' between these two events. Because the light pulse has a speed of ' c ', the time takes the pulse to travel from Jane to the mirror and back can be found from the definition of speed:

$$t_0 = \frac{\text{distance}}{\text{speed}} = \frac{2d}{c}$$

The time interval ' t_0 ' measured by Jane who is at rest in the moving vehicle requires only a single clock located at the **same place** in the frame.

A light beam seen by an observer **outside** the train

- Now consider the same pair of events viewed by David at O in a stationary frame as the figure below:



- According to David, the mirror and laser are moving to right with speed v

- By the time the light pulse reaches the mirror, the mirror moves a distance $\frac{vt_v}{2}$ where t_v is the time it takes the light to travel from O' to the mirror and back to O'' as measured by David.

Results:

- The light must travel **further** for David than it does for Jane. According to the second postulate of Special Relativity, the speed of light must be c as measured by **both observers**.
- Since the light travels a further distance for David, it follows that the time interval t_v measured by David in the stationary frame is longer than the time interval t_0 measured by Jane in the moving frame.
- Hence, according to David, Jane's clock (the moving clock compared to David) **runs slow**.

Time dilation equation:

$$t_v = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Dilated Time
The two events happen at different places

Proper time
The two events happen at the same place

t_0 is called **proper time**. It is the time interval between two events as **measured by an observer who see the event occur at the same place**. Jane measured proper time.

ANALYSING TIME DILATION QUESTIONS

- 1) Identify which observer measures proper time and which observer measures dilated time
- 2) Identify the data provided in the question as the proper time or the dilated time.
- 3) Use the time dilation equation to solve for any unknowns

EXAMPLE QUESTION 1

Sandra is travelling on a train and observes the second hand of her watch sweep through 15.0s. Jason is outside the train and also observes Sandra's watch as the train moves past him.

Sandra measured proper time as she is in the same frame of reference as her watch. Jason measures dilated time since he is moving relative to Sandra's watch. The 15.0s measured by Sandra is **proper time**.

$$t_v = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{15}{\sqrt{1 - \frac{(0.5c)^2}{c^2}}} = 17.35s$$

LENGTH CONTRACTION

The length of an object measured in a reference frame in which the object is moving is **always less** than the proper length. This effect is known as length contraction.

WHAT IS PROPER LENGTH

The **proper length** of an object is defined as the **length** of the object **measured in the reference frame in which the object is at rest**. For example, to measure the proper length of a moving object, you would need to be in the same reference frame as the moving object.

INTERPRETATION OF THE LENGTH CONTRACTION EQUATION

$$L_v = L_0 \sqrt{1 - \frac{v^2}{c^2}}$$

- It relates the **proper length** L_0 of an object as measured within its rest frame to the **contracted length** L_v as measured from a different frame of reference with a velocity v relative to the rest frame.
- $L_v < L_0$

The **length contraction takes place only along the direction of motion**. It is a symmetrical effect.

EVIDENCE OF TIME DILATION AND LENGTH CONTRACTION

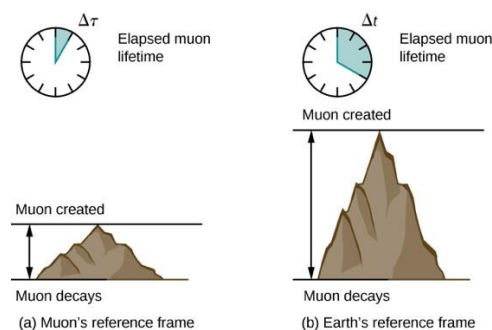
MUONS

Muons are unstable elementary particles, which have a charge equal to that of an electron and a mass 207 times greater. Muons can be produced by the absorption of cosmic radiation in the atmosphere.

- These unstable particles have a lifetime of 2.0 microseconds when measured in a reference frame at rest with them. After this time, they decay to produce energy and other particles.
- If we take 2.0 microseconds as the average lifetime of a muon and assume that their speed is close to the speed of light ($v = 0.998c$) we would find that these particles could only travel a **distance of only about 600m** before they decay into something else.
- Hence, they **cannot** reach the Earth from the upper atmosphere where they are produced. However, experiments show that a large number of muons **do reach the Earth**. The phenomenon of **time dilation** explains this effect.

Frame of Reference	Time taken for the muon to fall	Length of fall
Muon	Proper Time	Contracted length
Observer on Earth	Dilated time	Proper length

- In the muon's reference frame, there is no time dilation, but the distance travelled is observed to be shorter when measured in this frame.
- Likewise, in the Earth observer's reference frame, there is **time dilation**, but the distance of travel is measured to be the **proper height** of the mountain.
- In one frame the distance is shorter (length contraction) but the time is also shorter (proper time). In the other frame the distance is longer (proper length), but the time is also longer (dilated time)
- Proper length and proper time do not need to be measured by the same observer.



The Hafele-Keating involved **flying atomic clocks** on **aeroplanes** to **directly measure time dilation**.

3 atomic clocks were used. They all started at the United States Naval Observatory in Washington DC.

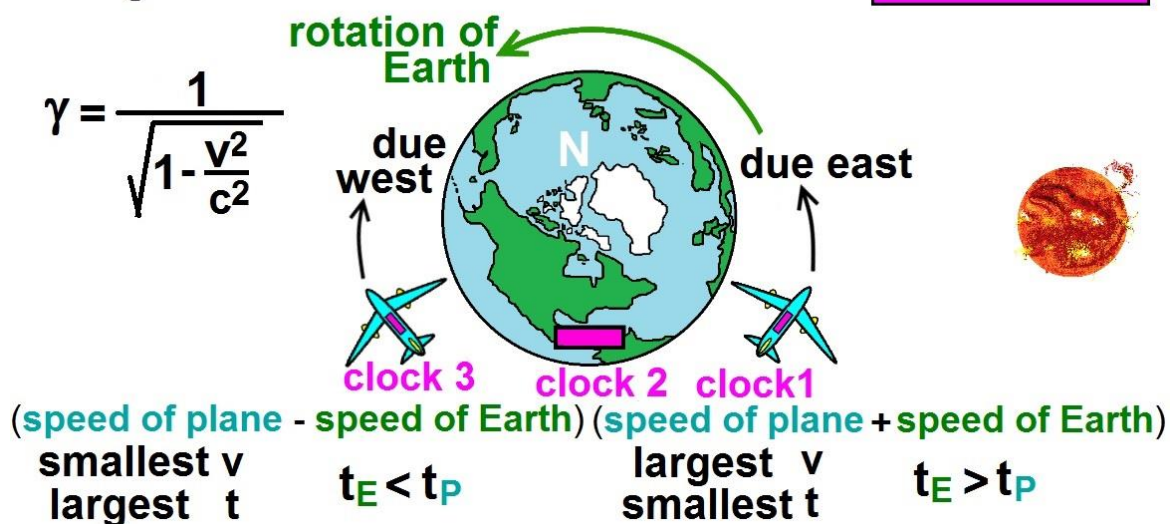
- 1) One clock was **left on the ground**
- 2) One clock was **flown around the world eastward**
- 3) One clock was **flown around the world westward**.

Once all the clocks returned to the same place, the time shown on each clock was compared:

- The times on each clock were **different** as each clock represented a **different amount of time dilation**.
- Atomic clocks have high precision so can measure small differences in time.

The Hafele-Keating Experiment of 1971

Using a series of HP 5061A cesium beam **atomic clock**



The experiment is best explained in the **Earth's axis reference frame**:

- The eastward clock has to move **faster** around the Earth's axis compared to the ground clock. Hence it experiences **more** time dilation.
- The eastward clock will record a shorter time than the clock on the ground (i.e. time dilated) because it is moving faster relative to the Earth's axis.
- The westward moving clock has to move **slower** around the Earth's axis compared to the ground clock. Hence it experiences **less** time dilation. It will record a longer time as it is moving slower.

None of the 3 atomic clocks measure proper time. This is because proper time is measured in the frame of reference of the Earth's axis. All three are moving in that frame. (But each clock measures proper time in its own frame of reference).

Particle accelerators use electric and magnetic fields to accelerate charged particles like protons, electrons and the nuclei of atoms to very high speeds approaching the speed of light. They can also produce particles that are otherwise unstable, which then go on to decay.

Evidence for Special Relativity can be seen in the **increased lifetime of short-lived particles in particle accelerators**.

- Unstable particles (like muons) are produced travelling at high speeds.
- Due to the high speed of the particles, their lifetime is **observed to be longer** in the Earth's reference frame, since **an observer in the Earth's frame would measure dilated time**.
- **This is the same as the muon phenomena.**

Evidence for length contraction is harder to observed due to the small size of the particles being accelerated. However, indirect evidence of length contraction can be observed in particle accelerators.

- The exact details of how particles behave in an accelerator at high speeds can only be explained if we assume they undergo length contraction – **increasing their density** and changing their shape of their electric field.
- The success of Special Relativity in explaining these results are evidence for length contraction in particle accelerators.

RELATIVISTIC MOMENTUM AND MASS DILATION

OUTCOMES

- describe the consequences and applications of relativistic momentum with reference to:
 - $p_v = \frac{m_0 v}{\sqrt{1 - \frac{v^2}{c^2}}}$
 - the limitation on the maximum velocity of a particle imposed by special relativity (ACSPH133)
- Use Einstein's mass–energy equivalence relationship $E = mc^2$ to calculate the energy released by processes in which mass is converted to energy, for example: (ACSPH134)
 - production of energy by the sun
 - particle–antiparticle interactions, eg positron–electron annihilation
 - combustion of conventional fuel

RELATIVISTIC MOMENTUM

Relativistic momentum can be derived from Newton's Second Law: $F = ma$.

- Using the definition of acceleration, $a = \frac{\Delta v}{\Delta t}$, it can be expressed in terms of momentum:

$$F = ma = m \left(\frac{\Delta v}{\Delta t} \right) = \frac{\Delta ma}{\Delta t} = \frac{\Delta p}{\Delta t}$$

- Newton's Second Law must be the same in all inertial frames (Einstein's 1st postulate and Galilean relativity)

All observers must agree on the meaning of force, and hence Newton's Second Law.

- This requires the ratio $\frac{\Delta p}{\Delta t}$ to be constant for all observers
- Since the definition of force in terms of momentum relies on time, we must take **time dilation** into account when changing reference frames.

$$F_{proper\ time} = \frac{p}{t_0} = \frac{m_0 v}{t_0}$$

$$F_{relativistic} = \frac{p_v}{t_v} = \frac{p_v \sqrt{1 - \frac{v^2}{c^2}}}{t_0}$$

$$\therefore \frac{m_0 v}{t_0} = \frac{p_v \sqrt{1 - \frac{v^2}{c^2}}}{t_0} \Rightarrow p_v = \frac{m_0 v}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Relativistic momentum is always: $p_v > p_0$

MASS DILATION

An object in motion will have a **greater mass** than when it is stationary

- This phenomenon is known as **mass dilation**
- This means that mass is also a **relative quantity**
- Mass dilation is a **consequence of relativistic momentum**

An observer that is **stationary** compared to the object will measure the **rest mass m_0** . Hence the momentum they measure will be $p_0 = m_0 v$

The observer moving at speed v compared to the object will measure relativistic momentum

$$p_v = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}} \times v$$

$$p_v = m_v \times v$$

- If we treat momentum as (mass x velocity) then we conclude that the moving observer measures a different mass. This mass is called **dilated mass** or **relativistic mass m_v**

$$m_v = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

- v is the relative velocity of the object and the observer, so has the same value for both
- **Relativistic momentum increases with velocity because mass increases with velocity**
- Mass is a measure of inertia – hence inertia increases with velocity.

LIMITING VELOCITY

The **limiting speed** of any object with **rest mass** is the speed of light. This is because as its speed approaches that of light, its mass dilates infinitely. It would therefore require an infinite amount of work to overcome this infinite inertia.

If an object is travelling near the speed of light, **doing work on it** will not increase its speed, but will **instead increase its mass**. This means that **mass is a form of energy**.

ENERGY

CONSERVATION OF MASS AND ENERGY

The **Law of Conservation of Energy** states that energy cannot be created or destroyed, but only **transformed from one type to another**.

Since **rest mass** is a type of energy, we can also transform it:

- Mass can be converted into other types of energy (mass reduces)
- Other types of energy can be converted into mass (mass increases)

Since mass is a type of energy and can be transformed to other forms of energy, **rest mass is no longer required to be conserved**. Conservation of Energy includes rest mass, since it is a type of energy. So it can be considered as the Conservation of Mass and Energy together, referred to as **mass-energy**.

Total relativistic mass of a particle is proportional to its total energy.

$$m_{relativistic} = \frac{1}{c^2}(m_0c^2 + K + U)$$

TIMELINE (APPROX)

- 1632: Galileo's proposed his Principle of Relativity
- 1650: Otto von Guericke invented a vacuum pump that allowed low air pressure to be achieved.
- 1675: Ole Romer and Christiaan Huygens. Speed of light by observing the motion of Io.
- 1678: Huygens' Wavelet Principle of Light
- 1692: Newton's Particle Model of Light
- 1801: Thomas Young's double slit experiment
- 1809: Etienne-Louis Malus quantifies the Malus's Law
- 1822: Augustin-Jean Fresnel presented a modified version of Huygens' model in which he proposed that light is a transverse wave
- 1849: Hippolyte Fizeau time of flight experiment
- 1855: Swiss school teacher Johann Jacob Balmer found an empirical formula that correctly predicted the wavelengths of four visible emission lines of hydrogen.
- 1862: Gustav Kirchhoff introduces black body
- 1869: Johann Hittorf found cathode rays
- 1869: Maltese cross experiment by Julius Plücker
- 1879: Paddle wheel experiment and deflection by magnetic fields experiment by William Crookes
- 1879: Thomson's charge-to-mass experiment
- 1873: James Clerk Maxwell proposed his equations
- 1887: Hertz wave experiment
- 1887: Michelson-Morley experiment
- 1850: Leon Foucault speed of light in water experiment
- 1897: Electron discovered by Thomson
- 1899: Alpha particle discovered by Rutherford
- 1901: Max Planck discovered a formula for blackbody radiation
- 1909: Robert A. Millikan and Harvey Fletcher attempted to measure the charge of an electron. (Millikan's Oil Drop Experiment)
- 1911: Thomson's plum pudding model of the atom
- 1911: Geiger-Marsden "Gold Foil experiment"
- 1913: Niels Bohr's quantum & classical model of the atom
- 1917: Rutherford conducted an experiment that proved hydrogen nuclei were present in heavier nuclei
- 1920: Rutherford named the 'fundamental particle' proton
- 1923: De Broglie's matter wave hypothesis
- 1926: Schrodinger's equation
- 1927: Davisson-Germer experiment
- 1932: James Chadwick was able to prove the existence of the neutron
- 1932: Positron discovered by Anderson
- 1932: De Sitter double-star experiment
- 1937: Muon discovered by Neddermeyer
- 1956: Electron antineutrino discovered by Reines and Cowan
- 1971: Hafele-Keating experiment

Model of the Atom

- Dalton
- Thomson
- Rutherford
- James Chadwick
- Bohr
- De Broglie